

ENGINEERING MATHEMATICS - I

UNIT - 1 : MATRICES

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By Gulshan sir

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UNIT-1 : Matrices

- Inverse of a matrix by Elementary transformations (L-1)
- Rank of matrix
 - (i) By determinant method (L-2)
 - (ii) By echelon form method (L-3)
 - (iii) By normal form method (L-4)
 - (iv) By normal form method (PAQ Form) (L-5)
- Solution of system of linear equations
 - (i) Non Homogenous System of linear equations (L-6)
 - (ii) Homogenous System of linear equations (L-7)
- Linear Dependence and Independence of vectors (L-8)
- Complex Matrices (L-9)
 - (i) Hermitian
 - (ii) Skew-Hermitian
 - (iii) Unitary Matrices
- Eigen values and Eigen vectors (L-10 + L-11)
- Cayley Hamilton Theorem (L-12)
- Applications to Engineering problems (L-12)

ENGINEERING MATHEMATICS - I

UNIT - 1 : MATRICES

Lecture - 1

Today's target

- Inverse of a matrix by Elementary transformation
- PYQs
- DPP



By Gulshan sir

Elementary Row Transformation

(i) Interchanging of any two rows

$$A = \begin{bmatrix} 3 & 1 & 4 \\ 2 & -3 & 4 \\ 1 & -1 & 1 \end{bmatrix} \begin{matrix} R_1 \\ R_2 \\ R_3 \end{matrix}$$

$$R_1 \longleftrightarrow R_3$$

$$A \sim \begin{bmatrix} 1 & -1 & 1 \\ 2 & -3 & 4 \\ 3 & 1 & 4 \end{bmatrix}$$

Elementary Column Transformation

(i) Interchanging of any two columns

$$A = \begin{bmatrix} 3 & 1 & 4 \\ 2 & -3 & 4 \\ 1 & -1 & 1 \end{bmatrix} \begin{matrix} C_1 & C_2 & C_3 \end{matrix}$$

$$C_1 \longleftrightarrow C_2$$

$$A \sim \begin{bmatrix} 1 & 3 & 4 \\ -3 & 2 & 4 \\ -1 & 1 & 1 \end{bmatrix}$$

(ii) Multiplication/Division of a row by a non-zero number

$$A = \begin{bmatrix} 1 & 4 & 8 \\ 4 & -2 & 4 \\ 3 & -1 & 1 \end{bmatrix} \begin{matrix} R_1 \\ R_2 \\ R_3 \end{matrix}$$

$$R_2 \rightarrow -\frac{1}{2} R_2$$

$$A \sim \begin{bmatrix} 1 & 4 & 8 \\ -2 & 1 & -2 \\ 3 & -1 & 1 \end{bmatrix}$$

(ii) Multiplication/Division of a column by a non-zero number

$$A = \begin{bmatrix} 1 & 5 & 6 \\ 3 & -3 & 4 \\ 5 & -1 & 2 \end{bmatrix} \begin{matrix} C_1 & C_2 & C_3 \end{matrix}$$

$$C_3 \rightarrow \frac{1}{2} C_3$$

$$A \sim \begin{bmatrix} 1 & 5 & 3 \\ 3 & -3 & 2 \\ 5 & -1 & 1 \end{bmatrix}$$

(iii) Addition of k times the element of a row to corresponding elements of another row, $k \neq 0$

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 1 & -3 & 4 \\ 5 & -1 & 1 \end{bmatrix} \begin{matrix} R_1 \\ R_2 \\ R_3 \end{matrix}$$

$$R_2 \rightarrow R_2 + (-1R_1)$$

$$R_3 \rightarrow R_3 + (-5R_1)$$

$$A \sim \begin{bmatrix} 1 & 2 & 4 \\ 0 & -5 & 0 \\ 0 & -11 & -19 \end{bmatrix}$$

(iii) Addition of k times the element of column to corresponding elements of another column, $k \neq 0$

$$A = \begin{bmatrix} 1 & 3 & 1 \\ 2 & -3 & 4 \\ 4 & -1 & 4 \end{bmatrix} \begin{matrix} C_1 & C_2 & C_3 \end{matrix}$$

$$C_2 \rightarrow C_2 + (-3C_1)$$

$$C_3 \rightarrow C_3 + (-1C_1)$$

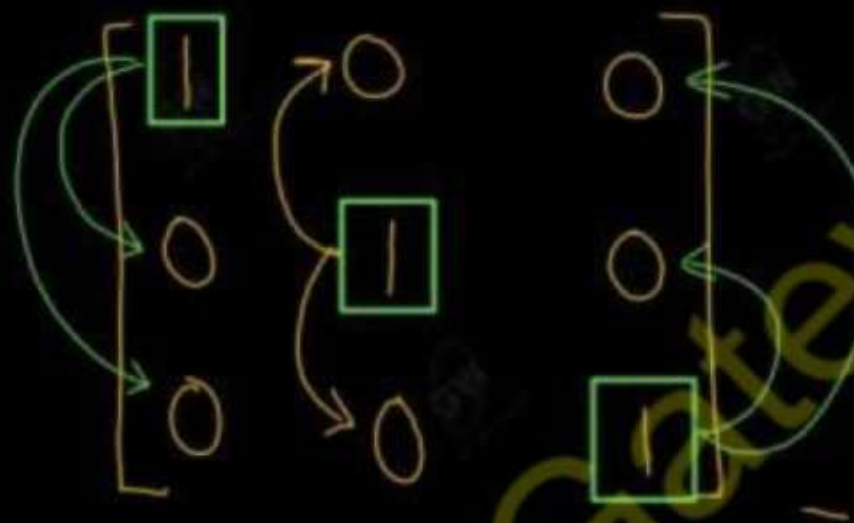
$$A \sim \begin{bmatrix} 1 & 0 & 0 \\ 2 & -19 & 2 \\ 4 & -13 & 0 \end{bmatrix}$$

Using elementary transformation

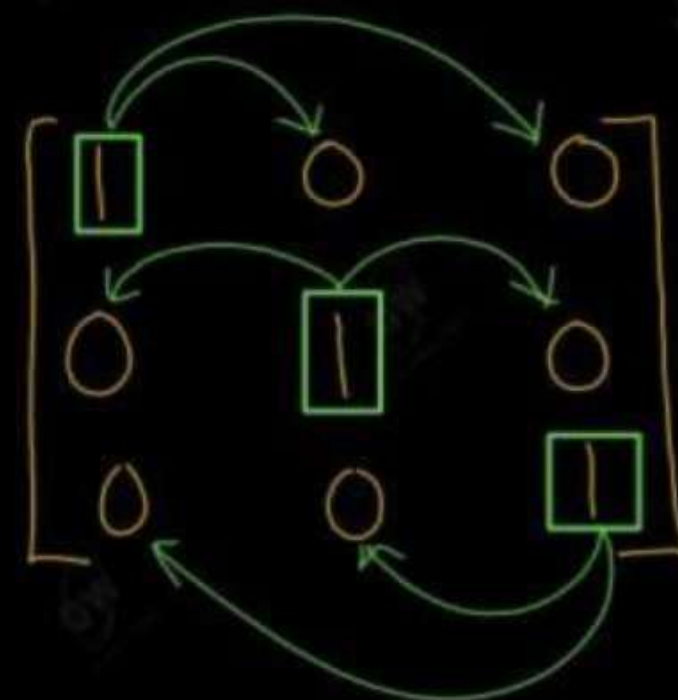
- (i) We can reduced a given matrix(square) in to Identity matrix
- (ii) We can reduced a given matrix in to Echelon form
- (iii) We can reduced a given matrix in to Normal form

Reduced a given matrix (square) in to Identity matrix

(i) Using elementary row transformation



(i) Using elementary column transformation



To find inverse of a Matrix

(i) Apply Elementary row transformation only

$$\begin{array}{l} A = IA \\ \rightarrow I = A^{-1}A \end{array}$$

(ii) Apply Elementary column transformation only

$$\begin{array}{l} A = AI \\ \rightarrow I = AA^{-1} \end{array}$$

NOTE :

(1) Don't apply both transformation in the same question

(2) if on applying one or more elementary row operation, we obtain all zeros in one or more rows, then

A^{-1} does not exist

Q.1 Find the inverse by employing elementary row transformation

$$A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$$

AKTU 2019

$$A = IA$$

$$\begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

$$R_1 \rightarrow R_1 + (-1R_2)$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

$$R_2 \rightarrow R_2 + (-2R_1)$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

$$R_2 \leftrightarrow R_3$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 1 \\ 0 & -3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 1 \\ -2 & 3 & 0 \end{bmatrix} A$$

$$R_2 \rightarrow -1R_2$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & \boxed{1} & -1 \\ 0 & -3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & -1 \\ -2 & 3 & 0 \end{bmatrix} A$$

$$R_3 \rightarrow R_3 + 3R_2$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & \boxed{1} \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & -1 \\ -2 & 3 & -3 \end{bmatrix} A$$

$$R_2 \rightarrow R_2 + R_3$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & -4 \\ -2 & 3 & -3 \end{bmatrix} A$$

$$I = A^{-1} A$$

$$A^{-1} = \begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & -4 \\ -2 & 3 & -3 \end{bmatrix}$$

Q.2 Employing elementary row transformations, find the inverse of the matrix

$$A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}$$

AKTU 2018

$$A = IA$$

$$\begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

$$R_1 \leftrightarrow R_2$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 3 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

$$R_3 \rightarrow R_3 + (-3R_1)$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & -5 & -8 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & -3 & 1 \end{bmatrix} A$$

$$R_1 \rightarrow R_1 + (-2R_2)$$

$$R_3 \rightarrow R_3 + 5R_2$$

$$\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} -2 & 1 & 0 \\ 1 & 0 & 0 \\ 5 & -3 & 1 \end{bmatrix} A$$

$$R_3 \rightarrow \frac{1}{2} R_3$$

$$\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & \boxed{1} \end{bmatrix} = \begin{bmatrix} -2 & 1 & 0 \\ 1 & 0 & 0 \\ -\frac{5}{2} & -\frac{3}{2} & \frac{1}{2} \end{bmatrix} A$$

$$R_1 \rightarrow R_1 + R_3$$

$$R_2 \rightarrow R_2 + (-2R_3)$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ -4 & 3 & -1 \\ \frac{5}{2} & -\frac{3}{2} & \frac{1}{2} \end{bmatrix} A$$

$$I = A^{-1}A$$

$$A^{-1} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ -4 & 3 & -1 \\ \frac{5}{2} & -\frac{3}{2} & \frac{1}{2} \end{bmatrix}$$

Q.3 Find the inverse of the following matrix using elementary row transformation

$$A = \begin{bmatrix} 2 & 3 & 4 \\ 4 & 3 & 1 \\ 1 & 2 & 4 \end{bmatrix}$$

AKTU 2021

$$A = IA$$

$$\begin{bmatrix} 2 & 3 & 4 \\ 4 & 3 & 1 \\ 1 & 2 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

$$R_1 \longleftrightarrow R_3$$

$$\begin{bmatrix} 1 & 2 & 4 \\ 4 & 3 & 1 \\ 2 & 3 & 4 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} A$$

$$R_2 \longrightarrow R_2 + (-4R_1)$$

$$R_3 \longrightarrow R_3 + (-2R_1)$$

$$\begin{bmatrix} 1 & 2 & 4 \\ 0 & -5 & -15 \\ 0 & -1 & -4 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & -4 \\ 1 & 0 & -2 \end{bmatrix}$$

$$R_2 \longleftrightarrow R_3$$

$$\begin{bmatrix} 1 & 2 & 4 \\ 0 & -1 & -4 \\ 0 & -5 & -15 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & -2 \\ 0 & 1 & -4 \end{bmatrix}$$

$$R_2 \longrightarrow -1R_2$$

$$\begin{bmatrix} 1 & 2 & 4 \\ 0 & 1 & 4 \\ 0 & -5 & -15 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ -1 & 0 & 2 \\ 0 & 1 & -4 \end{bmatrix} A$$

$$R_1 \rightarrow R_1 + (-2R_2)$$

$$R_3 \rightarrow R_3 + 5R_2$$

$$\begin{bmatrix} 1 & 0 & -4 \\ 0 & 1 & 4 \\ 0 & 0 & 5 \end{bmatrix} = \begin{bmatrix} 2 & 0 & -3 \\ -1 & 0 & 2 \\ -5 & 1 & 6 \end{bmatrix} A$$

$$R_3 \rightarrow \frac{1}{5} R_3$$

$$\begin{bmatrix} 1 & 0 & -4 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 & -3 \\ -1 & 0 & 2 \\ -1 & 1/5 & 6/5 \end{bmatrix}$$

$$R_1 \rightarrow R_1 + 4R_3$$

$$R_2 \rightarrow R_2 + (-4R_3)$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 4/5 & 9/5 \\ 3 & -4/5 & -14/5 \\ -1 & 1/5 & 6/5 \end{bmatrix} A$$

$$I = A^{-1} A$$

$$A^{-1} = \begin{bmatrix} -2 & \frac{4}{5} & \frac{9}{5} \\ 3 & -\frac{4}{5} & -\frac{14}{5} \\ -1 & \frac{1}{5} & \frac{6}{5} \end{bmatrix}$$

$$A^{-1} = \frac{1}{5} \begin{bmatrix} -10 & 4 & 9 \\ 15 & -4 & -14 \\ -5 & 1 & 6 \end{bmatrix}$$



Q.4 Find the inverse of the matrix by using elementary column transformation $A = \begin{bmatrix} 2 & 0 & -1 \\ 5 & 1 & 0 \\ 0 & 1 & 3 \end{bmatrix}$

$$A = A I$$

$$\begin{bmatrix} 2 & 0 & -1 \\ 5 & 1 & 0 \\ 0 & 1 & 3 \end{bmatrix} = A \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 5 & 1 & 5 \\ 3 & 1 & 6 \end{bmatrix} = A \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 2 \end{bmatrix}$$

$$C_1 \rightarrow C_1 + (-5C_2)$$

$$C_1 \rightarrow C_1 + C_3$$

$$C_3 \rightarrow C_3 + (-5C_2)$$

$$\begin{bmatrix} 1 & 0 & -1 \\ 5 & 1 & 0 \\ 3 & 1 & 3 \end{bmatrix} = A \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ -5 & 1 & -5 \\ 1 & 0 & 2 \end{bmatrix}$$

$$C_3 \rightarrow C_3 + C_1$$

$$C_1 \rightarrow C_1 + 2C_3$$

$$C_2 \rightarrow C_2 + (-1)C_3$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = A^{-1} \begin{bmatrix} 3 & -1 & 1 \\ -15 & 6 & -5 \\ 5 & -2 & 2 \end{bmatrix}$$

$$I = A A^{-1}$$

$$A^{-1} = \begin{bmatrix} 3 & -1 & 1 \\ -15 & 6 & -5 \\ 5 & -2 & 2 \end{bmatrix}$$

Inverse of a matrix by Elementary Transformation

Q.1 Transform $\begin{bmatrix} 1 & 3 & 3 \\ 2 & 4 & 10 \\ 3 & 8 & 4 \end{bmatrix}$ into a unit matrix by using elementary row transformation

Q.2 Find the inverse of the matrix by using elementary row transformation $A = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 2 & 3 \\ 1 & 1 & 2 \end{bmatrix}$

Ans $A^{-1} = \begin{bmatrix} -1/4 & 3/4 & -1 \\ 3/4 & -1/4 & 0 \\ -1/4 & -1/4 & 1 \end{bmatrix}$

Q.3 Employing elementary row transformations, find the inverse of the matrix $A = \begin{bmatrix} 8 & 4 & 3 \\ 2 & 1 & 1 \\ 1 & 2 & 1 \end{bmatrix}$

Ans $A^{-1} = \frac{1}{3} \begin{bmatrix} 1 & -2 & -1 \\ 1 & -5 & 2 \\ -3 & 12 & 0 \end{bmatrix}$

Q.4 Employing elementary row transformations, find the inverse of the following non-singular matrix

$$A = \begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 3 & 3 & 2 \\ 2 & 4 & 3 & 3 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

Ans $A^{-1} = \begin{bmatrix} 1 & -2 & 1 & 0 \\ 1 & -2 & 2 & -3 \\ 0 & 1 & -1 & 1 \\ -2 & 3 & -2 & 3 \end{bmatrix}$

ENGINEERING MATHEMATICS - I

UNIT - 1 : MATRICES

Lecture - 2

Today's target

- Rank of a matrix by Determinant method
- PYQs
- DPP



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Rank of a matrix by Determinant Method

Rank = Order of highest non zero determinant obtained from the matrix.

(P)

Note:

(i) If A is a null matrix, then

$$P(A) = 0$$

(ii) If A is NOT a null matrix, then

$$P(A) \geq 1$$

(iii) If A is non singular matrix of order $n \times n$, then

$$\rightarrow |A| \neq 0 \quad P(A) = n$$

(iv) If A is $m \times n$ matrix then

$$P(A) \leq \text{Minimum value of } m \text{ or } n$$

3×3

$$|A| \neq 0$$

$$P(A) = 3$$

$$|A| = 0$$

2×2

$$|A| \neq 0$$

$$P(A) = 2$$

$$|A| = 0$$

$$P(A) = 1$$

Q.1 Find the rank of the matrix $A = \begin{bmatrix} 2 & 4 \\ 4 & 5 \end{bmatrix}$.

$$A = \begin{bmatrix} 2 & 4 \\ 4 & 5 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 2 & 4 \\ 4 & 5 \end{vmatrix}$$

$$= 2 \times 5 - 4 \times 4$$

$$= 10 - 16$$

$$|A| = -6$$

$$\therefore |A| \neq 0$$

$$\rho(A) = 2$$

Q.2 Find the rank of the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \end{bmatrix}$.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \end{bmatrix}_{3 \times 3}$$

$$|D_1| = \begin{vmatrix} 1 & 2 \\ 2 & 4 \end{vmatrix}$$

$$|D_1| = 1 \times 4 - 2 \times 2$$

$$|D_1| = 0$$

$$|D_2| = \begin{vmatrix} 2 & 3 \\ 4 & 5 \end{vmatrix}$$

$$|D_2| = 2 \times 5 - 4 \times 3$$

$$|D_2| = 10 - 12$$

$$|D_2| = -2$$

$$|D_2| \neq 0$$

$$\rho(A) = 2$$

Q.3 Find the rank of the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 4 & 2 \\ 2 & 6 & 5 \end{bmatrix}$.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 4 & 2 \\ 2 & 6 & 5 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 1 & 2 & 3 \\ 1 & 4 & 2 \\ 2 & 6 & 5 \end{vmatrix}$$

$$\begin{aligned} |A| &= 1(20 - 12) - 2(5 - 4) + 3(6 - 8) \\ &= 8 - 2 - 6 \\ &= 8 - 8 \end{aligned}$$

$$|A| = 0 \quad \rho(A) \neq 3$$

$$\begin{aligned} |D_1| &= \begin{vmatrix} 1 & 2 \\ 1 & 4 \end{vmatrix} \\ &= 4 - 2 \end{aligned}$$

$$|D_1| = 2$$

$$|D_1| \neq 0$$

$$\rho(A) = 2$$

Q.4 Find the value of 'b' so that rank of $A = \begin{bmatrix} 2 & 4 & 2 \\ 3 & 1 & 2 \\ 1 & 0 & b \end{bmatrix}$ is 2.

$$A = \begin{bmatrix} 2 & 4 & 2 \\ 3 & 1 & 2 \\ 1 & 0 & b \end{bmatrix}$$

$$\rho(A) = 2$$

$$\therefore |A| = 0$$

$$\begin{vmatrix} 2 & 4 & 2 \\ 3 & 1 & 2 \\ 1 & 0 & b \end{vmatrix} = 0$$

$$2(b-0) - 4(3b-2) + 2(0-1) = 0$$

$$2b - 12b + 8 - 2 = 0$$

$$-10b + 6 = 0$$

$$+10b = +6$$

$$b = \frac{6}{10} = \frac{3}{5}$$

$$b = \frac{3}{5}$$

Q.5 Find the value of 'P' so that rank of $A = \begin{bmatrix} 3 & P & P \\ P & 3 & P \\ P & P & 3 \end{bmatrix}$ is 1.

$$A = \begin{bmatrix} 3 & P & P \\ P & 3 & P \\ P & P & 3 \end{bmatrix}$$

$$\rho(A) = 1$$

$$\begin{vmatrix} 3 & P \\ P & 3 \end{vmatrix} = 0$$

$$9 - P^2 = 0$$

$$P^2 = 9$$

$$P = \pm 3$$

$$\begin{vmatrix} P & P \\ 3 & P \end{vmatrix} = 0$$

$$P^2 - 3P = 0$$

$$P(P-3) = 0$$

$$P = 0$$

$$P = 3$$

$$P = 3$$

Q.6 Under what condition the rank of the following matrix is 3. $A = \begin{bmatrix} 2 & 4 & 2 \\ 2 & 1 & 2 \\ 1 & 0 & x \end{bmatrix}$

$$A = \begin{bmatrix} 2 & 4 & 2 \\ 2 & 1 & 2 \\ 1 & 0 & x \end{bmatrix}$$

$$\rho(A) = 3$$

$$|A| \neq 0$$

$$\begin{vmatrix} 2 & 4 & 2 \\ 2 & 1 & 2 \\ 1 & 0 & x \end{vmatrix} \neq 0$$

$$2(x-0) - 4(2x-2) + 2(0-1) \neq 0$$

$$2x - 8x + 8 - 2 \neq 0$$

$$-6x + 6 \neq 0$$

$$+6x \neq +6$$

$$x \neq 1$$

Topic : Rank of a matrix by Determinant Method

Q.1 Find the rank of the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 4 & 2 \\ 2 & 6 & 5 \end{bmatrix}$.

Ans.1 Rank = 3

Q.2 Find the value of 'K' so that rank of $A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & -1 & 2 \\ 4 & 1 & K \end{bmatrix}$ is 2.

Ans.2 $K = 5$

ENGINEERING MATHEMATICS - I

UNIT - 1 : MATRICES

Lecture - 3

Today's target

- Rank of a matrix by Echelon Form method
- PYQs
- DPP



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Rank of a matrix by Echelon Form method

STEP-1 : Convert the given matrix in to the Echelon form

STEP-2 : Rank (ρ) = Number of **non-zero rows** in the Echelon form

$$A = \begin{bmatrix} \boxed{1} & 3 & 2 & 4 \\ 0 & \boxed{1} & 0 & 2 \\ 0 & 0 & \boxed{3} & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{array}{l} \text{non zero row} \\ \text{non-zero row} \\ \text{non-zero row} \\ \text{zero row} \end{array}$$

$$\rho(A) = 3$$

Echelon Form

Any matrix A is said to be in Echelon form or row Echelon form if

- (i) The number of zeros before the first non-zero element increases from top to bottom

Note : First non zero element in each row is called Pivot element or Leading element

- (ii) If the column contain pivot element then all the entries below pivot element are zero.

Note : (a) Convert Pivot element in to 1, if require for easy calculation

(b) Pivot element shifted toward right

- (iii) All zero rows of the matrix lie below non zero row.

Note : The Echelon Form of a square matrix is an upper triangular matrix

Rank Nullity Theorem : If A is a matrix of order $m \times n$ then

$$\text{Rank of } A + \text{Nullity of } A = \text{Number of columns } (n)$$

How to convert a given matrix in to Echelon Form

$$(1) A = \begin{bmatrix} 2 & 7 \\ 3 & 10 \end{bmatrix}$$

$$(2) A = \begin{bmatrix} 2 & 1 & 5 \\ 7 & 2 & 0 \end{bmatrix}$$

$$(3) A = \begin{bmatrix} 1 & 5 \\ 4 & 2 \\ 6 & 4 \end{bmatrix}$$

$$(4) A = \begin{bmatrix} 2 & 3 & 4 \\ 4 & 3 & 1 \\ 1 & 2 & 4 \end{bmatrix}$$

$$(5) A = \begin{bmatrix} 1 & 8 & 5 & 1 \\ 4 & 6 & 1 & 4 \\ 7 & 4 & 2 & 5 \end{bmatrix}$$

$$(6) A = \begin{bmatrix} 2 & 6 & 3 \\ 4 & 4 & 4 \\ 5 & 5 & 8 \\ 5 & 2 & 9 \end{bmatrix}$$

$$(7) A = \begin{bmatrix} 1 & 1 & 2 & 3 \\ 4 & 1 & 0 & 2 \\ 2 & 3 & 0 & 4 \\ 0 & 1 & 0 & 2 \end{bmatrix}$$

Q.1 Find the rank of the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 4 & 2 \\ 2 & 6 & 5 \end{bmatrix}$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 4 & 2 \\ 2 & 6 & 5 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - R_1$$

$$R_3 \rightarrow R_3 + (-2R_1)$$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & -1 \\ 0 & 2 & -1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2$$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

It is an Echelon form

$\rho(A)$ = Number of non-zero rows

$$\rho(A) = 2$$

Q.2 Find the rank of the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$ by Echelon Form

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$$

$$R_2 \rightarrow R_2 + (-2R_1)$$

$$R_3 \rightarrow R_3 + (-3R_1)$$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

It is an Echelon Form

$\rho(A)$ = Number of non-zero row

$$\rho(A) = 1$$

Q.3 Find the rank of the matrix A =

$$A = \begin{bmatrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{bmatrix}$$

AKTU 2019

$$A = \begin{bmatrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - R_1$$

$$R_3 \rightarrow R_3 - R_1$$

$$A = \begin{bmatrix} 2 & 2 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

It is an Echelon Form

$\rho(A)$ = Number of non-zero rows

$$\rho(A) = 1$$

Q.4 Find the rank of the matrix $A = \begin{bmatrix} 1 & 2 & 3 & 2 \\ 2 & 3 & 5 & 1 \\ 1 & 3 & 4 & 5 \end{bmatrix}$ Also find nullity

$$A = \begin{bmatrix} 1 & 2 & 3 & 2 \\ 2 & 3 & 5 & 1 \\ 1 & 3 & 4 & 5 \end{bmatrix}$$

$$R_2 \rightarrow R_2 + (-2R_1)$$

$$R_3 \rightarrow R_3 - R_1$$

$$A = \begin{bmatrix} 1 & 2 & 3 & 2 \\ 0 & -1 & -1 & -3 \\ 0 & 1 & 1 & 3 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + R_2$$

$$A = \begin{bmatrix} 1 & 2 & 3 & 2 \\ 0 & -1 & -1 & -3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$\rho(A)$ = Number of non-zero row

$$\rho(A) = 2$$

$$\begin{aligned} \text{Nullity} &= \text{No. of column} - \text{Rank of matrix} \\ &= 4 - 2 \\ &= 2 \end{aligned}$$

Q.5 Find the rank of the matrix $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 9 \\ 3 & 6 & 10 & 13 \end{bmatrix}$

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 9 \\ 3 & 6 & 10 & 13 \end{bmatrix}$$

$$R_2 \rightarrow R_2 + (-2R_1)$$

$$R_3 \rightarrow R_3 + (-3R_1)$$

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

$$R_2 \leftrightarrow R_3$$

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$\rho(A)$ = Number of non-zero rows

$$\rho(A) = 3$$

Q.6 Use elementary transformation to reduce the following matrix A to triangular form and hence,

$$A = \begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$$

$$R_1 \leftrightarrow R_2$$

$$A = \begin{bmatrix} 1 & -1 & -2 & -4 \\ 2 & 3 & -1 & -1 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$$

$$R_2 \rightarrow R_2 + (-2R_1)$$

$$R_3 \rightarrow R_3 + (-3R_1)$$

Find the rank of $A = \begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$

$$R_4 \rightarrow R_4 + (-6R_1)$$

$$A = \begin{bmatrix} 1 & -1 & -2 & -4 \\ 0 & 5 & 3 & 7 \\ 0 & 4 & 9 & 10 \\ 0 & 9 & 12 & 17 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - R_3$$

$$A = \begin{bmatrix} 1 & -1 & -2 & -4 \\ 0 & 1 & -6 & -3 \\ 0 & 4 & 9 & 10 \\ 0 & 9 & 12 & 17 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + (-4R_2)$$

$$R_4 \rightarrow R_4 + (-9R_2)$$

$$A = \begin{bmatrix} 1 & -1 & -2 & -4 \\ 0 & 1 & -6 & -3 \\ 0 & 0 & 33 & 22 \\ 0 & 0 & 66 & 44 \end{bmatrix}$$

$$R_4 \rightarrow R_4 + (-2R_3)$$

$$A = \begin{bmatrix} 1 & -1 & -2 & -1 \\ 0 & 1 & -6 & -3 \\ 0 & 0 & 33 & 22 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$\rho(A)$ = Number of non-zero row

$$\rho(A) = 3$$

Topic : Rank of a matrix by Echelon Form method

Q 1 Find the rank of the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \\ 3 & 6 & 10 \end{bmatrix}$

Ans 1 Rank = 2

Q 2 Find the rank of the matrix $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 1 & 2 \end{bmatrix}$

Ans 2 Rank = 3

Q 3 Use elementary transformation to reduce the following matrix A to triangular form and hence

find the rank of $A = \begin{bmatrix} -1 & 2 & 3 & -2 \\ 2 & -5 & 1 & 2 \\ 3 & -8 & 5 & 2 \\ 5 & -12 & -1 & 6 \end{bmatrix}$

Ans 3 Rank = 2

ENGINEERING MATHEMATICS - I

UNIT - 1 : MATRICES

Lecture - 4

Today's target

- Rank of a matrix by Normal Form method
- PYQs
- DPP



By Gulshan sir

- Reduce given matrix in to Normal form using both elementary row transformation and column transformation
- Matrix can be reduced in to one of the following Normal form

(i) I_r

(ii) $\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$

(iii) $[I_r \ 0]$

(iv) $\begin{bmatrix} I_r \\ 0 \end{bmatrix}$

Where r = Rank of the matrix

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Q.1 Reduce the matrix $\begin{bmatrix} 1 & 1 & 1 \\ 3 & 1 & 1 \end{bmatrix}$ in to normal form and find its rank

AKTU 2022

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 3 & 1 & 1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 + (-3R_1)$$

$$\sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & -2 & -2 \end{bmatrix}$$

$$C_2 \rightarrow C_2 - C_1$$

$$C_3 \rightarrow C_3 - C_1$$

$$\sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & -2 \end{bmatrix}$$

$$R_2 \rightarrow -\frac{1}{2}R_2$$

$$\sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

$$C_3 \rightarrow C_3 - C_2$$

$$\sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\sim \begin{bmatrix} I_2 & 0 \end{bmatrix}$$

Normal Form

$$\rho(A) = 2$$

Q.2 Reduce the matrix A to its normal form and hence find its rank where A = $\begin{bmatrix} 3 & 2 & -1 \\ 4 & 2 & 6 \\ 7 & 4 & 5 \end{bmatrix}$ AKTU 2016

$$A = \begin{bmatrix} 3 & 2 & -1 \\ 4 & 2 & 6 \\ 7 & 4 & 5 \end{bmatrix}$$

$$C_1 \rightarrow C_1 - C_2$$

$$\sim \begin{bmatrix} 1 & 2 & -1 \\ 2 & 2 & 6 \\ 3 & 4 & 5 \end{bmatrix}$$

$$R_2 \rightarrow R_2 + (-2R_1)$$

$$R_3 \rightarrow R_3 + (-3R_1)$$

$$\sim \begin{bmatrix} 1 & 2 & -1 \\ 0 & -2 & 8 \\ 0 & -2 & 8 \end{bmatrix}$$

$$C_2 \rightarrow C_2 + (-2C_1)$$

$$C_3 \rightarrow C_3 + C_1$$

$$\sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 8 \\ 0 & -2 & 8 \end{bmatrix}$$

$$C_2 \rightarrow -\frac{1}{2}C_2$$

$$\sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 8 \\ 0 & 1 & 8 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2$$

$$\sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 8 \\ 0 & 0 & 0 \end{bmatrix}$$

$$C_3 \rightarrow C_3 + (-8C_2)$$

$$\sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\sim \begin{bmatrix} I_2 & 0 \\ 0 & 0 \end{bmatrix}$$

Normal Form

$$\rho(A) = 2$$

Reduce the matrix A to its normal form and hence find its rank where $A = \begin{bmatrix} 1 & 2 & 1 & 0 \\ -2 & 4 & 3 & 0 \\ 1 & 0 & 2 & -8 \end{bmatrix}$

AKTU 2015

$$A = \begin{bmatrix} 1 & 2 & 1 & 0 \\ -2 & 4 & 3 & 0 \\ 1 & 0 & 2 & -8 \end{bmatrix}$$

$$R_2 \rightarrow R_2 + 2R_1$$

$$R_3 \rightarrow R_3 - R_1$$

$$\sim \begin{bmatrix} 1 & 2 & 1 & 0 \\ 0 & 8 & 5 & 0 \\ 0 & -2 & 1 & -8 \end{bmatrix}$$

$$C_2 \rightarrow C_2 + (-2C_1)$$

$$C_3 \rightarrow C_3 - C_1$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 8 & 5 & 0 \\ 0 & -2 & 1 & -8 \end{bmatrix}$$

$$R_2 \leftrightarrow R_3$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -2 & 1 & -8 \\ 0 & 8 & 5 & 0 \end{bmatrix}$$

$$C_2 \rightarrow -\frac{1}{2}C_2$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & -8 \\ 0 & -4 & 5 & 0 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + 4R_2$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & -8 \\ 0 & 0 & 9 & -32 \end{bmatrix}$$

$$C_3 \rightarrow C_3 - C_2$$

$$C_4 \rightarrow C_4 + 8C_2$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 9 & -32 \end{bmatrix}$$

$$C_3 \rightarrow \frac{1}{9} C_3$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -32 \end{bmatrix}$$

$$C_4 \rightarrow C_4 + 32 C_3$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$\sim \begin{bmatrix} I_3 & 0 \end{bmatrix}$$

Normal Form

$$\rho(A) = 3$$

Q 4 Reduce the matrix A to its normal form and hence find its rank where

$$A = \begin{bmatrix} -2 & -1 & 3 & -1 \\ 1 & 2 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \end{bmatrix}$$

AKTU 2018

$$A = \begin{bmatrix} -2 & -1 & 3 & -1 \\ 1 & 2 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \end{bmatrix}$$

$$R_1 \leftrightarrow R_2$$

$$\sim \begin{bmatrix} 1 & 2 & -3 & -1 \\ -2 & -1 & 3 & -1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 + 2R_1$$

$$R_3 \rightarrow R_3 - R_1$$

$$\sim \begin{bmatrix} 1 & 2 & -3 & -1 \\ 0 & 3 & -3 & -3 \\ 0 & -2 & 4 & 2 \\ 0 & 1 & 1 & -1 \end{bmatrix}$$

$$C_2 \rightarrow C_2 + (-2)C_1$$

$$C_3 \rightarrow C_3 + 3C_1$$

$$C_4 \rightarrow C_4 + C_1$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 3 & -3 & -3 \\ 0 & -2 & 4 & 2 \\ 0 & 1 & 1 & -1 \end{bmatrix}$$

$$R_2 \leftrightarrow R_4$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & -1 \\ 0 & -2 & 4 & 2 \\ 0 & 3 & -3 & -3 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + 2R_2$$

$$R_4 \rightarrow R_4 + (-3R_2)$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 6 & 0 \\ 0 & 0 & -6 & 0 \end{bmatrix}$$

$$C_3 \rightarrow C_3 - C_2$$

$$C_4 \rightarrow C_4 + C_2$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 6 & 0 \\ 0 & 0 & -6 & 0 \end{bmatrix}$$

$$C_3 \rightarrow \frac{1}{6} C_3$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$R_4 \rightarrow R_4 - R_3$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\sim \begin{bmatrix} I_3 & 0 \\ 0 & 0 \end{bmatrix}$$

Normal
Form

$$p(A) = 3$$

Q 5 Reduce the matrix A to its normal form and hence find its rank where

$$A = \begin{bmatrix} 1 & 2 & -1 & 4 \\ 2 & 4 & 3 & 4 \\ 1 & 2 & 3 & 4 \\ -1 & -2 & 6 & -7 \end{bmatrix}$$

AKTU 2019

$$A = \begin{bmatrix} 1 & 2 & -1 & 4 \\ 2 & 4 & 3 & 4 \\ 1 & 2 & 3 & 4 \\ -1 & -2 & 6 & -7 \end{bmatrix}$$

$$R_2 \rightarrow R_2 + (-2R_1)$$

$$R_3 \rightarrow R_3 - R_1$$

$$R_4 \rightarrow R_4 + R_1$$

$$\sim \begin{bmatrix} 1 & 2 & -1 & 4 \\ 0 & 0 & 5 & -4 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 5 & -3 \end{bmatrix}$$

$$C_2 \rightarrow C_2 + (-2C_1)$$

$$C_3 \rightarrow C_3 + C_1$$

$$C_4 \rightarrow C_4 + (-4C_1)$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 5 & -4 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 5 & -3 \end{bmatrix}$$

$$C_2 \leftrightarrow C_3$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 5 & 0 & -4 \\ 0 & 4 & 0 & 0 \\ 0 & 5 & 0 & -3 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - R_3$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -4 \\ 0 & 4 & 0 & 0 \\ 0 & 5 & 0 & -3 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + (-4R_2)$$

$$R_4 \rightarrow R_4 + (-5R_2)$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -4 \\ 0 & 0 & 0 & 16 \\ 0 & 0 & 0 & 17 \end{bmatrix}$$

$$C_4 \rightarrow C_4 + 4C_2$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 16 \\ 0 & 0 & 0 & 17 \end{bmatrix}$$

$$C_3 \leftrightarrow C_4$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 16 & 0 \\ 0 & 0 & 17 & 0 \end{bmatrix}$$

$$R_3 \rightarrow \frac{1}{16} R_3$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 17 & 0 \end{bmatrix}$$

$$R_4 \rightarrow R_4 + (-17R_3)$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\sim \begin{bmatrix} I_3 & 0 \\ 0 & 0 \end{bmatrix} \text{ Normal Form}$$

$$\rho(A) = 3$$

Q 6 Reduce the matrix A to its normal form and hence find its rank where

$$A = \begin{bmatrix} 1 & 3 & 4 & 2 \\ 2 & -1 & 3 & 2 \\ 3 & -5 & 2 & 2 \\ 6 & -3 & 8 & 6 \end{bmatrix}$$

AKTU 2020

$$A = \begin{bmatrix} 1 & 3 & 4 & 2 \\ 2 & -1 & 3 & 2 \\ 3 & -5 & 2 & 2 \\ 6 & -3 & 8 & 6 \end{bmatrix}$$

$$R_2 \rightarrow R_2 + (-2R_1)$$

$$R_3 \rightarrow R_3 + (-3R_1)$$

$$R_4 \rightarrow R_4 + (-6R_1)$$

$$\sim \begin{bmatrix} 1 & 3 & 4 & 2 \\ 0 & -7 & -5 & -2 \\ 0 & -14 & -10 & -4 \\ 0 & -21 & -16 & -6 \end{bmatrix}$$

$$C_2 \rightarrow C_2 + (-3C_1)$$

$$C_3 \rightarrow C_3 + (-4C_1)$$

$$C_4 \rightarrow C_4 + (-2C_1)$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -7 & -5 & -2 \\ 0 & -14 & -10 & -4 \\ 0 & -21 & -16 & -6 \end{bmatrix}$$

$$C_2 \rightarrow -\frac{1}{7}C_2$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -5 & -2 \\ 0 & 2 & -10 & -4 \\ 0 & 3 & -16 & -6 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + (-2R_2)$$

$$R_4 \rightarrow R_4 + (-3R_2)$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -5 & -2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix}$$

$$C_3 \rightarrow C_3 + 5C_2$$

$$C_4 \rightarrow C_4 + 2C_2$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix}$$

$$R_3 \leftrightarrow R_4$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$R_3 \rightarrow -R_3$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\sim \begin{bmatrix} I_3 & 0 \\ 0 & 0 \end{bmatrix}$$

Normal Form

$$P(A) = 3$$

Topic : Rank of a matrix by Normal Form method

Q.1 Reduce the matrix A to its normal form and hence find its rank where $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \\ 3 & 6 & 10 \end{bmatrix}$

Ans.1 Rank = 3

Q.2 Reduce the matrix A to its normal form and hence find its rank where $A = \begin{bmatrix} -1 & -3 & 3 & -1 \\ 1 & 1 & -1 & 0 \\ 2 & -5 & 2 & -3 \end{bmatrix}$ AKTU 2014

Ans.1 Rank = 3

Q.3 Reduce the matrix A to its normal form and hence find its rank where $A = \begin{bmatrix} 0 & 1 & 2 & -1 \\ 1 & 2 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$ AKTU 2017

Ans.3 Rank = 4

Q.4 Reduce the matrix A to its normal form and hence find its rank where $A = \begin{bmatrix} 1 & -1 & 2 & -3 \\ 4 & 1 & 0 & 2 \\ 0 & 3 & 0 & 4 \\ 0 & 1 & 0 & 2 \end{bmatrix}$

Ans.4 Rank = 4

ENGINEERING MATHEMATICS - I

UNIT -1 : MATRICES

Lecture - 5

Today's target

- Rank of a matrix by Normal Form method (PAQ Form)
- PYQs
- DPP



By Gulshan sir

Rank of a matrix by Normal Form (PAQ Form)

Both row and column elementary transformation are allowed

$$A_{m \times n} = I_m A I_n$$

$$\text{Normal Form} = PAQ$$

- (i) Reduce the matrix A on LHS in normal form by using row and column transformation
- (ii) Row transformation is also applied on I_m (Pre factor)
- (iii) Column transformation is also applied on I_n (Post factor)

NOTE : P and Q are not unique

$$AA^{-1} = I$$

Inverse a matrix by Normal Form (PAQ Form)

$$I = PAQ$$

$$P^{-1}I = P^{-1}PAQ$$

$$P^{-1} = IAQ$$

$$P^{-1}P = AQP$$

$$I = AQP$$

$$A^{-1}I = A^{-1}AQP$$

$$A^{-1} = QP$$

Q.1 If $A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$ Determine two non-singular matrices P and Q such that $PAQ = I$.

$$A_{3 \times 3} = I_3 A I_3$$

$$\begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_1 \rightarrow R_1 - R_2$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 + (-2R_1)$$

Hence find A^{-1}

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix} \xrightarrow{C_2 \rightarrow C_2 + C_3} \begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{C_3 \rightarrow C_3 + (-4C_2)} \begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

$$C_3 \rightarrow C_3 + (-4C_2)$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -4 \\ 0 & 1 & -3 \end{bmatrix}$$

$$I_3 = PAQ$$

$$\rho(A) = 3$$

$$P = \begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$Q = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -4 \\ 0 & 1 & -3 \end{bmatrix}$$

$$A^{-1} = QP$$

$$A^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -4 \\ 0 & 1 & -3 \end{bmatrix} \begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & -4 \\ -2 & 3 & -3 \end{bmatrix}$$

Q.2 Find non-singular matrices P and Q such that PAQ is in the normal form for the matrix hence find

the rank of matrix $A = \begin{bmatrix} 1 & 2 & 3 & -2 \\ 2 & -2 & 1 & 3 \\ 3 & 0 & 4 & 1 \end{bmatrix}$

AKTU 2015

$$A_{3 \times 4} = I_3 A I_4$$

$$\begin{bmatrix} 1 & 2 & 3 & -2 \\ 2 & -2 & 1 & 3 \\ 3 & 0 & 4 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 + (-2R_1) \quad R_3 \rightarrow R_3 + (-3R_1)$$

$$\begin{bmatrix} 1 & 2 & 3 & -2 \\ 0 & -6 & -5 & 7 \\ 0 & -6 & -5 & 7 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$C_2 \rightarrow C_2 + (-2C_1)$$

$$C_3 \rightarrow C_3 + (-3C_1)$$

$$C_4 \rightarrow C_4 + 2C_1$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -6 & -5 & 7 \\ 0 & -6 & -5 & 7 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & -2 & -3 & 2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$C_2 \rightarrow C_2 + C_4$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -5 & 7 \\ 0 & 1 & -5 & 7 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 0 & -3 & 2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -5 & 7 \\ 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -1 & -1 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 0 & -3 & 2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

$$C_3 \rightarrow C_3 + 5C_2 \quad C_4 \rightarrow C_4 + (-7)C_2$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -1 & -1 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 0 & -3 & 2 \\ 0 & 1 & 5 & -7 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 5 & -6 \end{bmatrix}$$

$$\begin{bmatrix} I_2 & 0 \\ 0 & 0 \end{bmatrix} = PAQ$$

$$\rho(A) = 2$$

$$P = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -1 & -1 & 1 \end{bmatrix} \checkmark$$

$$Q = \begin{bmatrix} 1 & 0 & -3 & 2 \\ 0 & 1 & 5 & -7 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 5 & -6 \end{bmatrix} \checkmark$$

Topic : Rank of a matrix by Normal Form method (PAQ Form)

Q.1 For the matrix $A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 0 & -1 & -1 \end{bmatrix}$ Find non-singular matrices P and Q such that PAQ is a normal form.

AKTU 2021

Ans.1 $P = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}$ and $Q = \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$

Q.2 Find non-singular matrices P and Q such that PAQ is a normal form for the matrix $A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 0 & -1 & -1 \end{bmatrix}$

Ans.2 $P = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}, Q = \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$

ENGINEERING MATHEMATICS - I

UNIT -1 : MATRICES

Lecture - 6

Today's target

- A system of **Non-Homogenous** Linear Equations
- PYQs
- DPP



By Gulshan sir

System of Linear Equations

Non-Homogenous System of Linear Equations

$$\underline{a_1x} + \underline{b_1y} + \underline{c_1z} = d_1$$

$$\underline{a_2x} + \underline{b_2y} + \underline{c_2z} = d_2$$

$$\underline{a_3x} + \underline{b_3y} + \underline{c_3z} = d_3$$

Homogenous System of Linear Equations

$$\underline{a_1x} + \underline{b_1y} + \underline{c_1z} = 0$$

$$\underline{a_2x} + \underline{b_2y} + \underline{c_2z} = 0$$

$$\underline{a_3x} + \underline{b_3y} + \underline{c_3z} = 0$$

Gateway Classes

Non-Homogenous System of Linear Equations

$$a_1x + b_1y + c_1z = d_1$$

$$a_2x + b_2y + c_2z = d_2$$

$$a_3x + b_3y + c_3z = d_3$$

In Matrix Form

$$\begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$$

$$AX = B$$

$$AX = B$$

$$A^{-1}AX = A^{-1}B$$

$$IX = A^{-1}B$$

$$X = A^{-1}B$$

Augmented Matrix

$$[A:B] = \begin{bmatrix} a_1 & b_1 & c_1 & : & d_1 \\ a_2 & b_2 & c_2 & : & d_2 \\ a_3 & b_3 & c_3 & : & d_3 \end{bmatrix}$$

A System of non-Homogenous Linear Equations
 $AX = B$

Find $\rho(A)$ and $\rho[A: B]$

$\rho(A) = \rho[A: B]$

Solution exists, system
is consistent

$\rho(A) \neq \rho[A: B]$

No Solution, system
is inconsistent

$\rho(A) = \rho[A: B] = n(\text{no. of unknowns})$

Unique
Solution

$\rho(A) = \rho[A: B] < n(\text{no. of unknowns})$

Infinite
Solution

consider arbitrary constants
 $= n - r$

$r = \text{rank}$

$$x + y + z = 3, \quad x + 2y + 3z = 4, \quad x + 4y + 9z = 6.$$

$$x + y + z = 3$$

$$x + 2y + 3z = 4$$

$$x + 4y + 9z = 6$$

In Matrix Form

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 9 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \\ 6 \end{bmatrix}$$

$$AX = B$$

Augmented Matrix

$$[A : B] = \begin{bmatrix} 1 & 1 & 1 & : & 3 \\ 1 & 2 & 3 & : & 4 \\ 1 & 4 & 9 & : & 6 \end{bmatrix}$$

$$R_2 \rightarrow R_2 + (-R_1)$$

$$R_3 \rightarrow R_3 + (-R_1)$$

$$[A : B] = \begin{bmatrix} 1 & 1 & 1 & : & 3 \\ 0 & 1 & 2 & : & 1 \\ 0 & 3 & 8 & : & 3 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + (-3R_2)$$

$$[A : B] = \begin{bmatrix} 1 & 1 & 1 & : & 3 \\ 0 & 1 & 2 & : & 1 \\ 0 & 0 & 2 & : & 0 \end{bmatrix}$$

$$\rho(A) = 3$$

$$\rho(A : B) = 3$$

$$n = 3$$

$$\therefore \rho(A) = \rho(A:B) = n(\text{Unknown})$$

The given system is consistent and have unique solution

$$x + y + z = 3 \quad \text{--- (1)}$$

$$y + 2z = 1 \quad \text{--- (2)}$$

$$2z = 0 \quad \text{--- (3)}$$

$$z = 0$$

$$y = 1$$

$$x = 2$$

$$x + y + z = 6,$$

$$x + 2y + 3z = 14,$$

$$x + 4y + 7z = 30$$

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$$x + y + z = 6$$

$$x + 2y + 3z = 14$$

$$x + 4y + 7z = 30$$

Augmented Matrix

$$[A:B] = \begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 1 & 2 & 3 & : & 14 \\ 1 & 4 & 7 & : & 30 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - R_1$$

$$R_3 \rightarrow R_3 - R_1$$

$$[A:B] = \begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 0 & 1 & 2 & : & 8 \\ 0 & 3 & 6 & : & 24 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + (-3R_2)$$

$$[A:B] = \begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 0 & 1 & 2 & : & 8 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$$

$$\rho(A) = 2$$

$$\rho(A:B) = 2$$

$$n = 3$$

$$\rho(A) = \rho(A:B) < n$$

The given system is consist and have
infinite solutions

$$x + y + z = 6 \quad \text{--- ①}$$

$$y + 2z = 8 \quad \text{--- ②}$$

$$\text{Let } z = k$$

$$y + 2k = 8$$

$$y = 8 - 2k$$

$$x + 8 - 2k + k = 6$$

$$x + 8 - k = 6$$

$$x = k - 2$$

$$z = k$$

$$y = 8 - 2k$$

$$x = k - 2$$

Q.3 Show that the system of equations $x + y + z = -3$, $3x + y - 2z = -2$, $2x + 4y + 7z = 7$

is not consistent.

$$x + y + z = -3$$

$$3x + y - 2z = -2$$

$$2x + 4y + 7z = 7$$

Augmented Matrix

$$[A:B] = \begin{bmatrix} 1 & 1 & 1 & : & -3 \\ 3 & 1 & -2 & : & -2 \\ 2 & 4 & 7 & : & 7 \end{bmatrix}$$

$$R_2 \rightarrow R_2 + (-3R_1)$$

$$R_3 \rightarrow R_3 + (-2R_1)$$

$$[A:B] = \begin{bmatrix} 1 & 1 & 1 & : & -3 \\ 0 & -2 & -5 & : & 7 \\ 0 & 2 & 5 & : & 13 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + R_2$$

$$[A:B] = \begin{bmatrix} 1 & 1 & 1 & : & -3 \\ 0 & -2 & -5 & : & 7 \\ 0 & 0 & 0 & : & 20 \end{bmatrix}$$

$$\rho(A) = 2$$

$$\rho(A:B) = 3$$

$$\rho(A) \neq \rho(A:B)$$

The given system is inconsistent and has no solution

$$x + y + z = 6, \quad x + 2y + 3z = 10, \quad x + 2y + \lambda z = \mu$$

have (i) no solution (ii) unique solution (iii) infinite solutions?

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$$x + y + z = 6$$

$$x + 2y + 3z = 10$$

$$x + 2y + \lambda z = \mu$$

Augmented matrix

$$[A:B] = \begin{bmatrix} 1 & 1 & 1 & : & 6 \\ \textcircled{1} & 2 & 3 & : & 10 \\ \textcircled{1} & \textcircled{2} & \lambda & : & \mu \end{bmatrix}$$

$$R_2 \rightarrow R_2 - R_1$$

$$R_3 \rightarrow R_3 - R_1$$

$$[A:B] = \begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 0 & 1 & 2 & : & 4 \\ 0 & \textcircled{1} & \lambda - 1 & : & \mu - 6 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2$$

$$[A:B] = \begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 0 & 1 & 2 & : & 4 \\ 0 & 0 & \lambda - 3 & : & \mu - 10 \end{bmatrix}$$

(i) For no solution

$$\lambda - 3 = 0$$

$$\boxed{\lambda = 3}$$

$$\mu - 10 \neq 0$$

$$\boxed{\mu \neq 10}$$

(II) Unique solution

$$\lambda - 3 \neq 0$$

$$\lambda \neq 3$$

μ can have any value

(III) Infinite solution

$$\lambda - 3 = 0$$

$$\lambda = 3$$

$$\mu - 10 = 0$$

$$\mu = 10$$

Q.5 Apply the matrix method to solve the system of equations

$$x_1 - 2x_2 + x_3 - x_4 + 1 = 0$$

$$3x_1 - 2x_3 + 3x_4 + 4 = 0$$

$$5x_1 - 4x_2 + x_4 + 3 = 0$$

$$x_1 - 2x_2 + x_3 - x_4 = -1$$

$$3x_1 + 0x_2 - 2x_3 + 3x_4 = -4$$

$$5x_1 - 4x_2 + 0x_3 + x_4 = -3$$

Augmented Matrix

$$[A:B] = \begin{bmatrix} 1 & -2 & 1 & -1 & : & -1 \\ \textcircled{3} & 0 & -2 & 3 & : & -4 \\ \textcircled{5} & -4 & 0 & 1 & : & -3 \end{bmatrix}$$

$$R_2 \rightarrow R_2 + (-3R_1)$$

$$R_3 \rightarrow R_3 + (-5R_1)$$

$$[A:B] = \begin{bmatrix} 1 & -2 & 1 & -1 & : & -1 \\ 0 & 6 & -5 & 6 & : & -1 \\ 0 & \textcircled{6} & -5 & 6 & : & 2 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2$$

$$[A:B] = \begin{bmatrix} 1 & -2 & 1 & -1 & : & -1 \\ 0 & 6 & -5 & 6 & : & -1 \\ 0 & 0 & 0 & 0 & : & 3 \end{bmatrix}$$

$$\rho(A) = 2$$

$$\rho(A:B) = 3$$

$$\rho(A) \neq \rho(A:B)$$

Given system

is inconsistent

and has no solution

Q.6 For what value of k , the equations $x + y + z = 1$, $2x + y + 4z = k$, $4x + y + 10z = k^2$

have a solution and solve them completely in each case.

$$x + y + z = 1$$

$$2x + y + 4z = k$$

$$4x + y + 10z = k^2$$

Augmented Matrix

$$[A:B] = \begin{bmatrix} 1 & 1 & 1 & : & 1 \\ 2 & 1 & 4 & : & k \\ 4 & 1 & 10 & : & k^2 \end{bmatrix}$$

$$R_2 \rightarrow R_2 + (-2R_1)$$

$$R_3 \rightarrow R_3 + (-4R_1)$$

$$[A:B] = \begin{bmatrix} 1 & 1 & 1 & : & 1 \\ 0 & -1 & 2 & : & k-2 \\ 0 & -3 & 6 & : & k^2-4 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + (-3R_2)$$

$$[A:B] = \begin{bmatrix} 1 & 1 & 1 & : & 1 \\ 0 & -1 & 2 & : & k-2 \\ 0 & 0 & 0 & : & k^2-3k+2 \end{bmatrix}$$

The given system has solution

$$K^2 - 3K + 2 = 0$$

$$(K-1)(K-2) = 0$$

$$K = 1, 2$$

CASE-1 ($K=1$)

$$x + y + z = 1 \quad \text{--- (1)}$$

$$-y + 2z = -1 \quad \text{--- (2)}$$

Let $z = a$

$$-y + 2a = -1$$

$$-y = -2a - 1$$

$$y = 2a + 1$$

From (1)

$$x + 2a + 1 + a = 1$$

$$x + 3a + 1 = 1$$

$$x = -3a$$

CASE-2 ($K=2$)

$$x + y + z = 1 \quad \text{--- (3)}$$

$$-y + 2z = 0 \quad \text{--- (4)}$$

Let $z = b$

$$-y + 2b = 0$$

$$y = 2b$$

$$x + 2b + b = 1$$

$$x = -3b + 1$$

Topic : Solution of Non-Homogenous System of Linear Equations

Q.1 Solve, the system of equations using matrix method:

$$2x_1 + x_2 + 2x_3 + x_4 = 6,$$

$$6x_1 - 6x_2 + 6x_3 + 12x_4 = 36,$$

Ans $x_1 = 2, x_2 = 1, x_3 = -1, x_4 = 3$

$$4x_1 + 3x_2 + 3x_3 - 3x_4 = -1,$$

$$2x_1 + 2x_2 - x_3 + x_4 = 10.$$

Q.2 Test for consistency and hence, solve the following set of equations :

$$10y + 3z = 0$$

$$3x + 3y + z = 1$$

$$2x - 3y - z = 5$$

$$x + 2y = 4$$

Ans Inconsistent, no solution exists

Q.3 Determine the values of λ and μ such that the system

$$2x - 5y + 2z = 8$$

$$2x + 4y + 6z = 5$$

$$x + 2y + \lambda z = \mu$$

have (i) no solution (ii) a unique solution (iii) infinite solutions

Ans (i) $\lambda = 3$ and $\mu \neq 5/2$ (ii) $\lambda \neq 3$ and μ may have any value (iii) $\lambda = 3$ and $\mu = 5/2$

Q.4 For what values of λ and μ , the system of linear equations

$$x + y + z = 6$$

$$x + 2y + 5z = 10$$

$$2x + 3y + \lambda z = \mu$$

has (i) a unique solution (ii) no solution (iii) infinite solutions also find the solution for $\lambda = 2$ and $\mu = 8$.

Ans.4 (i) $\lambda \neq 6$, μ is arbitrary (ii) $\lambda = 6$, $\mu \neq 16$

(iii) $\lambda = 6$, $\mu = 16$

when $\lambda = 2$, $\mu = 8$, solution is $x = 8$, $y = -4$ and $z = 2$.

Q.5 Show that the equations

$$-2x + y + z = a$$

$$x - 2y + z = b$$

$$x + y - 2z = c$$

Have no solution unless $a + b + c = 0$. In which case they have infinitely many solutions? Find these solution when $a = 1$, $b = 1$, $c = -2$

Ans

Q.6 Solve by calculating the inverse by elementary row operations:

$$x_1 + x_2 + x_3 + x_4 = 0$$

$$x_1 + x_2 + x_3 - x_4 = 4$$

$$x_1 + x_2 - x_3 + x_4 = -4$$

$$x_1 - x_2 + x_3 + x_4 = 2$$

$$X = A^{-1}B$$

Ans.5 $x_1 = 1$, $x_2 = -1$, $x_3 = 2$, $x_4 = -2$

AKTU 2020

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ENGINEERING MATHEMATICS - I

UNIT -1 : MATRICES

Lecture - 7

Today's target

- A system of **Homogenous** Linear Equations
- PYQs
- DPP



By Gulshan sir

System of Linear Equations

Non-Homogenous System of Linear Equations

$$\underline{a_1x} + \underline{b_1y} + \underline{c_1z} = \underline{d_1}$$

$$\underline{a_2x} + \underline{b_2y} + \underline{c_2z} = \underline{d_2}$$

$$\underline{a_3x} + \underline{b_3y} + \underline{c_3z} = \underline{d_3}$$

Homogenous System of Linear Equations

$$\underline{a_1x} + \underline{b_1y} + \underline{c_1z} = 0$$

$$\underline{a_2x} + \underline{b_2y} + \underline{c_2z} = 0$$

$$\underline{a_3x} + \underline{b_3y} + \underline{c_3z} = 0$$

Gateway Classes

Homogenous System of Linear Equations

$$a_1x + b_1y + c_1z = 0$$

$$a_2x + b_2y + c_2z = 0$$

$$a_3x + b_3y + c_3z = 0$$

In Matrix Form

$$\begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$AX = 0$$

$$A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$$

A System of Homogenous Linear Equations
 $AX = 0$

Always has a Solution
(or system is consistent)

Find $\rho(A)$

$\rho(A) = n(\text{no. of unknowns})$

$\rho(A) < n(\text{no. of unknowns})$

Unique or trivial
Solution

Zero Solution

Infinite no. of non trivial
Solution

Non-Zero Solution

consider Arbitrary constants = $n - r$

Q.1 Test whether the following system of equations possess a non-trivial solution:

$$x_1 + x_2 + 2x_3 + 3x_4 = 0$$

$$5x_1 + 7x_2 + 11x_3 + 17x_4 = 0$$

$$3x_1 + 4x_2 + 7x_3 + 10x_4 = 0$$

$$6x_1 + 8x_2 + 13x_3 + 16x_4 = 0.$$

$$x_1 + x_2 + 2x_3 + 3x_4 = 0$$

$$3x_1 + 4x_2 + 7x_3 + 10x_4 = 0$$

$$5x_1 + 7x_2 + 11x_3 + 17x_4 = 0$$

$$6x_1 + 8x_2 + 13x_3 + 16x_4 = 0$$

In Matrix Form

$$\begin{bmatrix} 1 & 1 & 2 & 3 \\ 3 & 4 & 7 & 10 \\ 5 & 7 & 11 & 17 \\ 6 & 8 & 13 & 16 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 1 & 2 & 3 \\ 3 & 4 & 7 & 10 \\ 5 & 7 & 11 & 17 \\ 6 & 8 & 13 & 16 \end{bmatrix}$$

$$R_2 \rightarrow R_2 + (-3R_1)$$

$$R_3 \rightarrow R_3 + (-5R_1)$$

$$R_4 \rightarrow R_4 + (-6R_1)$$

$$A \sim \begin{bmatrix} 1 & 1 & 2 & 3 \\ 0 & 1 & 1 & 1 \\ 0 & 2 & 1 & 2 \\ 0 & 2 & 1 & -2 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + (-2R_2)$$

$$R_4 \rightarrow R_4 + (-2R_2)$$

$$A \sim \begin{bmatrix} 1 & 1 & 2 & 3 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & -1 & -4 \end{bmatrix}$$

$$R_4 \rightarrow R_4 + (-R_3)$$

$$A \sim \begin{bmatrix} 1 & 1 & 2 & 3 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -4 \end{bmatrix}$$

Echelon Form

$$\rho(A) = 4 = \text{No. of Unknown}$$

No, The given system has unique solution or trivial solution or zero solution

$$x_1 = 0$$

$$x_2 = 0$$

$$x_3 = 0$$

$$x_4 = 0$$

Q.2 Solve the system of homogenous equations :

$$x_1 + x_2 + x_3 + x_4 = 0$$

$$x_1 + 3x_2 + 2x_3 + 4x_4 = 0$$

$$2x_1 + x_3 - x_4 = 0$$

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$$x_1 + x_2 + x_3 + x_4 = 0$$

$$x_1 + 3x_2 + 2x_3 + 4x_4 = 0$$

$$2x_1 + 0x_2 + x_3 - x_4 = 0$$

In Matrix Form

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 0 & 1 & -1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 + (-R_1)$$

$$R_3 \rightarrow R_3 + (-2R_1)$$

$$A \sim \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 1 & 3 \\ 0 & -2 & -1 & -3 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + R_2$$

$$A \sim \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\rho(A) = 2 < \text{No. of unknown}$$

The given system of equation has infinite non-trivial solution

$$x_1 + x_2 + x_3 + x_4 = 0 \quad \text{--- (1)}$$

$$2x_2 + x_3 + 3x_4 = 0 \quad \text{--- (2)}$$

Let $x_3 = k_1$

$$x_4 = k_2$$

Put x_3 and x_4 in (2)

$$2x_2 + k_1 + 3k_2 = 0$$

$$x_2 = -\frac{1}{2}(k_1 + 3k_2)$$

Put x_2 , x_3 and x_4 in (1)

$$x_1 = \frac{1}{2}(k_1 - k_2)$$

Q.3 Determine k such that the system of homogenous equations

$$2x + y + 2z = 0, \quad x + y + 3z = 0, \quad 4x + 3y + kz = 0$$

has (i) trivial solution (ii) non trivial solutions. Find the non trivial solutions using matrix method.

$$2x + y + 2z = 0$$

$$x + y + 3z = 0$$

$$4x + 3y + kz = 0$$

In Matrix Form

$$\begin{bmatrix} 2 & 1 & 2 \\ 1 & 1 & 3 \\ 4 & 3 & k \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$A = \begin{bmatrix} 2 & 1 & 2 \\ 1 & 1 & 3 \\ 4 & 3 & k \end{bmatrix}$$

$$R_1 \leftrightarrow R_2$$

$$A \sim \begin{bmatrix} 1 & 1 & 3 \\ 2 & 1 & 2 \\ 4 & 3 & k \end{bmatrix}$$

$$R_2 \rightarrow R_2 + (-2R_1)$$

$$R_3 \rightarrow R_3 + (-4R_1)$$

$$A \sim \begin{bmatrix} 1 & 1 & 3 \\ 0 & -1 & -4 \\ 0 & -1 & k-12 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + (-R_2)$$

$$A \sim \begin{bmatrix} 1 & 1 & 3 \\ 0 & -1 & -4 \\ 0 & 0 & k-8 \end{bmatrix}$$

Echelon Form

(i) For trivial solution

$$k - 8 \neq 0$$

$$k \neq 8$$

(ii) Non trivial solution

$$k - 8 = 0$$

$$k = 8$$

Put $k = 8$ in

Echelon Form

$$A \sim \begin{bmatrix} 1 & 1 & 3 \\ 0 & -1 & -4 \\ 0 & 0 & 0 \end{bmatrix}$$

$$x + y + 3z = 0 \quad \text{--- (1)}$$

$$-y - 4z = 0 \quad \text{--- (2)}$$

Let

$$z = k$$

$$y = -4k$$

$$x = k$$

Q.4 Find the values of λ for which the equations

$$(\lambda - 1)x + (3\lambda + 1)y + 2\lambda z = 0,$$

$$2x + (3\lambda + 1)y + 3(\lambda - 1)z = 0$$

have a non-trivial solution. Also find the solution in each case.

$$(\lambda - 1)x + (3\lambda + 1)y + 2\lambda z = 0$$

$$(\lambda - 1)x + (4\lambda - 2)y + (\lambda + 3)z = 0$$

$$2x + (3\lambda + 1)y + 3(\lambda - 1)z = 0$$

In Matrix Form

$$\begin{bmatrix} \lambda - 1 & 3\lambda + 1 & 2\lambda \\ \lambda - 1 & 4\lambda - 2 & \lambda + 3 \\ 2 & 3\lambda + 1 & 3(\lambda - 1) \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$(\lambda - 1)x + (4\lambda - 2)y + (\lambda + 3)z = 0$$

$$A = \begin{bmatrix} \lambda - 1 & 3\lambda + 1 & 2\lambda \\ \lambda - 1 & 4\lambda - 2 & \lambda + 3 \\ 2 & 3\lambda + 1 & 3\lambda - 3 \end{bmatrix}$$

For Non trivial solution

$$|A| = 0$$

$$\begin{vmatrix} \lambda - 1 & 3\lambda + 1 & 2\lambda \\ \lambda - 1 & 4\lambda - 2 & \lambda + 3 \\ 2 & 3\lambda + 1 & 3\lambda - 3 \end{vmatrix} = 0$$

$$C_1 \rightarrow C_1 + C_2 + C_3$$

$$\begin{vmatrix} 6\lambda & 3\lambda+1 & 2\lambda \\ 6\lambda & 4\lambda-2 & \lambda+3 \\ 6\lambda & 3\lambda+1 & 3\lambda-3 \end{vmatrix} = 0$$

Taking 6λ common from C_1

$$6\lambda \begin{vmatrix} 1 & 3\lambda+1 & 2\lambda \\ \textcircled{1} & 4\lambda-2 & \lambda+3 \\ \textcircled{1} & 3\lambda+1 & 3\lambda-3 \end{vmatrix} = 0$$

$$R_2 \rightarrow R_2 + (-R_1)$$

$$R_3 \rightarrow R_3 + (-R_1)$$

$$6\lambda \begin{vmatrix} 1 & 3\lambda+1 & 2\lambda \\ 0 & \lambda-3 & -\lambda+3 \\ 0 & 0 & \lambda-3 \end{vmatrix} = 0$$

Expand along C_1

$$6\lambda [(\lambda-3)(\lambda-3) - 0] = 0$$

$$6\lambda(\lambda-3)(\lambda-3) = 0$$

$$\lambda = 0, 3$$

CASE - I When $\lambda = 0$

$$A = \begin{bmatrix} -1 & 1 & 0 \\ \textcircled{-1} & -2 & 3 \\ \textcircled{2} & 1 & -3 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - R_1$$

$$R_3 \rightarrow R_3 + 2R_1$$

$$A \sim \begin{bmatrix} -1 & 1 & 0 \\ 0 & -3 & 3 \\ 0 & 3 & -3 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + R_2$$

$$A \sim \begin{bmatrix} -1 & 1 & 0 \\ 0 & -3 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

Echelon Form

$$\rho(A) = 2 < \text{No. of unknown}$$

Non-trivial solution

$$-x + y = 0 \quad \text{--- (1)}$$

$$-3y + 3z = 0 \quad \text{--- (2)}$$

$$\text{Let } z = k$$

From (1) and (2)

$$y = k$$

$$x = k$$

$$x = k$$

$$y = k$$

$$z = k$$

CASE-3 When $\lambda = 3$

$$A = \begin{bmatrix} 2 & 10 & 6 \\ 2 & 10 & 6 \\ 2 & 10 & 6 \end{bmatrix}$$

$$R_2 \rightarrow R_2 + (-R_1)$$

$$R_3 \rightarrow R_3 + (-R_1)$$

$$A \sim \begin{bmatrix} 2 & 10 & 6 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\rho(A) = 1 < \text{No. of Unknown}$$

Non-trivial solution

$$2x + 10y + 6z = 0 \quad \text{--- (1)}$$

$$\text{Let } y = k_1$$

$$z = k_2$$

Put y and z in (1)

$$2x + 10k_1 + 6k_2 = 0$$

$$x = -5k_1 - 3k_2$$

Topic : A system of Homogenous Linear Equations

Q.1 Solve the equations using matrix method:

$$x_1 + 3x_2 + 2x_3 = 0, \quad 2x_1 - x_2 + 3x_3 = 0, \quad 3x_1 - 5x_2 + 4x_3 = 0, \quad x_1 + 17x_2 + 4x_3 = 0. \quad \text{Ans non trivial}$$

Q.2 Find the value of k so that the equations : $x + y + 3z = 0$, $4x + 3y + kz = 0$, $2x + y + 2z = 0$

have a non trivial solutions.

Ans k = 8

Q.3 Find the values of k for which the system of equations:

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$$(3k - 8)x + 3y + 3z = 0$$

$$3x + (3k - 8)y + 3z = 0$$

$$3x + 3y + (3k - 8)z = 0 \quad \text{has a non-trivial solution.} \quad \text{Ans } K = 2/3, 11/3, 11/3$$

Q.4 Find the values of λ for which the equations

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$$x + (\lambda + 4)y + (4\lambda + 2)z = 0$$

$$x + 2(\lambda + 1)y + (3\lambda + 4)z = 0$$

$$2x + 3\lambda y + (3\lambda + 4)z = 0$$

have a non-trivial solution. Also find the solution in each case.

Ans $\lambda = 2, -2$

ENGINEERING MATHEMATICS - I

UNIT -1 : MATRICES

Lecture - 8

Today's target

- Linear Dependence and Linear Independence of Vectors
- PYQs
- DPP



By Gulshan sir

UNIT-1 : Matrices

- Inverse of a matrix by Elementary transformations (L-1)
- Rank of matrix
 - (i) By determinant method (L-2)
 - (ii) By echelon form method (L-3)
 - (iii) By normal form method (L-4)
 - (iv) By normal form method (PAQ Form) (L-5)
- Solution of system of linear equations
 - (i) Non Homogenous System of linear equations (L-6)
 - (ii) Homogenous System of linear equations (L-7)
- Linear Dependence and Independence of vectors (L-8)
- Complex Matrices (L-9)
 - (i) Hermitian
 - (ii) Skew-Hermitian
 - (iii) Unitary Matrices
- Eigen values and Eigen vectors (L-10 + L-11)
- Cayley Hamilton Theorem (L-12)
- Applications to Engineering problems (L-12)

WORKING RULE

Step – 1 : Construct coefficient matrix A with elements of given vectors as columns.

Step – 2 : Find $\rho(A)$

Step – 3 : (i) If $\rho(A) < \text{no. of vectors}$

Then the given set of vectors is **linearly dependent**.

A set of n Vectors(Matrices) $X_1, X_2, \dots, X_n$ is said to be linearly dependent if there exist n scalars

$k_1, k_2, k_3, \dots, k_n$, not all zero, such that

$$k_1 X_1 + k_2 X_2 + \dots + k_n X_n = 0$$

(ii) If $\rho(A) = \text{no. of vectors}$

Then the given set of vectors is **linearly independent**.

A set of n Vectors(Matrices) $X_1, X_2, \dots, X_n$ is said to be linearly independent if $k_1 = k_2 = k_3 = \dots = k_n = 0$

Q.1 Find whether or not the following vectors are linearly dependent or independent **(1, 1, 1, 1)**,

(0, 1, 1, 1), **(0, 0, 1, 1)**, **(0, 0, 0, 1)**.

Let

$$\checkmark X_1 = (1, 1, 1, 1)$$

$$\checkmark X_2 = (0, 1, 1, 1)$$

$$\checkmark X_3 = (0, 0, 1, 1)$$

$$\checkmark X_4 = (0, 0, 0, 1)$$

coefficient Matrix

$$A = \begin{bmatrix} \boxed{1} & 0 & 0 & 0 \\ \textcircled{1} & 1 & 0 & 0 \\ \textcircled{1} & 1 & 1 & 0 \\ \textcircled{1} & 1 & 1 & 1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - R_1$$

$$R_3 \rightarrow R_3 - R_1$$

$$R_4 \rightarrow R_4 - R_1$$

$$A \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \boxed{1} & 0 & 0 \\ 0 & \textcircled{1} & 1 & 0 \\ 0 & \textcircled{1} & 1 & 1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2$$

$$R_4 \rightarrow R_4 - R_2$$

$$A \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \boxed{1} & 0 \\ 0 & 0 & \textcircled{1} & 1 \end{bmatrix}$$

$$R_4 \rightarrow R_4 - R_3$$

$$A \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

It is in Echelon Form

$$\rho(A) = 4 (= \text{No. of vectors})$$

Hence, the given set of vectors is linearly independent

Here

$$K_1 = K_2 = K_3 = K_4 = 0$$

Show that the column vectors of $A = \begin{bmatrix} 1 & 0 & 0 \\ 6 & 2 & 1 \\ 4 & 3 & 2 \end{bmatrix}$ are linearly independent.

OR

Show that the vectors $X_1 = (1, 6, 4)$, $X_2 = (0, 2, 3)$, $X_3 = (0, 1, 2)$ are linearly dependent. **AKTU 2020**

Given

$$X_1 = (1, 6, 4)$$

$$X_2 = (0, 2, 3)$$

$$X_3 = (0, 1, 2)$$

Coefficient Matrix

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 6 & 2 & 1 \\ 4 & 3 & 2 \end{bmatrix}$$

$$R_2 \rightarrow R_2 + (-6R_1)$$

$$R_3 \rightarrow R_3 + (-4R_1)$$

$$A \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 3 & 2 \end{bmatrix}$$

$$R_2 \rightarrow \frac{1}{2} R_2$$

$$A \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & \frac{1}{2} \\ 0 & 3 & 2 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + (-3R_2)$$

$$A \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & \frac{1}{2} \end{bmatrix}$$

Echelon Form

$$\rho(A) = 3 \text{ (= Number of vectors)}$$

Hence,

The given set of vectors is linearly independent

Gateway Classes

Q.3 Show that the vectors $X_1 = (1, -1, 1)$, $X_2 = (2, 1, 1)$, $X_3 = (3, 0, 2)$ are linearly dependent. Find

the relation between them.

AKTU 2013

$$X_1 = (1, -1, 1)$$

$$X_2 = (2, 1, 1)$$

$$X_3 = (3, 0, 2)$$

Coefficient Matrix

$$A = \begin{bmatrix} 1 & 2 & 3 \\ -1 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$$

$$R_2 \rightarrow R_2 + R_1$$

$$R_3 \rightarrow R_3 - R_1$$

$$A \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 3 & 3 \\ 0 & -1 & -1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + \frac{1}{3}R_2$$

$$A \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 3 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

Echelon Form

$$\rho(A) = 2 < \text{Number of vectors}$$

Hence

The given set of vectors is linearly dependent

Let the relation between X_1 , X_2 and X_3 is

$$K_1 X_1 + K_2 X_2 + K_3 X_3 = 0$$

①

k_1, k_2 and k_3 are calculated
from Echelon Form

$$k_1 + 2k_2 + 3k_3 = 0 \quad \text{--- (2)}$$

$$3k_2 + 3k_3 = 0 \quad \text{--- (3)}$$

Let $k_2 = k$

Put k_2 in (3)

$$3k + 3k_3 = 0$$

$$3k_3 = -3k$$

$$k_3 = -k$$

Put k_2 and k_3 in (1)

$$k_1 + 2k - 3k = 0$$

$$k_1 - k = 0$$

$$k_1 = k$$

Put k_1, k_2 and k_3
in (1)

$$kx_1 + kx_2 - kx_3 = 0$$

$$kx_1 + kx_2 = kx_3$$

$$\cancel{k}(x_1 + x_2) = \cancel{k}x_3$$

$$x_1 + x_2 = x_3$$

Q.4 Find the value of λ for which the vectors $(1, -2, \lambda)$, $(2, -1, 5)$, and $(3, -5, 7\lambda)$ are linearly dependent.

Let

$$x_1 = (1, -2, \lambda)$$

$$x_2 = (2, -1, 5)$$

$$x_3 = (3, -5, 7\lambda)$$

Coefficient Matrix

$$A = \begin{bmatrix} 1 & 2 & 3 \\ -2 & -1 & -5 \\ \lambda & 5 & 7\lambda \end{bmatrix}$$

$$\rho(A) < 3$$

$$\Rightarrow |A| = 0$$

$$\begin{vmatrix} 1 & 2 & 3 \\ -2 & -1 & -5 \\ \lambda & 5 & 7\lambda \end{vmatrix} = 0$$

Expand along R_1

$$1(-7\lambda + 25) - 2(-14\lambda + 5\lambda) + 3(-10 + \lambda) = 0$$

$$-7\lambda + 25 + 28\lambda - 10\lambda - 30 + 3\lambda = 0$$

$$28\lambda - 14\lambda - 5 = 0$$

$$14\lambda = 5$$

$$\lambda = \frac{5}{14}$$

Topic : Linear Dependence and Linear Independence of Vectors

Q.1 Examine the following vectors for linear dependence and find the relation between them, if possible:

$$X_1 = (1, 1, -1, 1), \quad X_2 = (1, -1, 2, -1), \quad X_3 = (3, 1, 0, 1)$$

Ans Linearly dependent; $2X_1 + X_2 = X_3 = 8$

Q.2 Show that the vectors $X_1 = (1, 2, 4)$, $X_2 = (2, -1, 3)$, $X_3 = (0, 1, 2)$ and $X_4 = (-3, 7, 2)$ are linearly dependent and find the relation between them.

Ans Linearly dependent; $9X_1 - 12X_2 + 5X_3 - 5X_4 = 0$

Q.3 Find whether or not the following vectors are linearly dependent or independent $(1, 1, 0)$, $(1, 0, 1)$, $(0, 1, 1)$.

Ans $\rho(A) = 2 (< 3, \text{ no of vectors})$, Hence the given set of vectors is linearly dependent.

Q.4 Show that row vectors of the matrix $\begin{bmatrix} 1 & 2 & -2 \\ -1 & 3 & 0 \\ 4 & -2 & 1 \end{bmatrix}$ are linearly independent

ENGINEERING MATHEMATICS - I

UNIT -1 : MATRICES

Lecture - 9

Today's target

➤ Complex Matrices

(i) Hermitian Matrix (ii) Skew-Hermitian Matrix (iii) Unitary Matrix

➤ PYQs

➤ DPP



By Gulshan sir

If at least one element of a matrix is a complex number $Z = a + ib$ where a, b are real and $i = \sqrt{-1}$, then the matrix is called complex matrix

Examples

(i) $A = \begin{bmatrix} 1 & -1 & 0 \\ 1 & 1 & i \\ 3 & 1 & 2 \end{bmatrix}$ ✓

(ii) $A = \begin{bmatrix} 2+i & 3 & -1+3i \\ -5 & i & 4-2i \end{bmatrix}$ ✓

(iii) $A = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 3 & 1 & 2 \end{bmatrix}$ ✗

Real Part

Imaginary Part

$$i = \sqrt{-1}$$

Conjugate of Complex Number ($\bar{Z}$)

$$\checkmark \quad Z = a + ib$$

$$\bar{Z} = a - ib$$

Conjugate of complex Matrix ($\bar{A}$)

$$A = \begin{bmatrix} 1 & -i & 1+i \\ i & 0 & 2-3i \\ 1-i & 2+3i & 2 \end{bmatrix}$$

$$\bar{A} = \begin{bmatrix} 1 & i & 1-i \\ -i & 0 & 2+3i \\ 1+i & 2-3i & 2 \end{bmatrix}$$

Note

$$(i) \quad \overline{\bar{A}} = A$$

$$(ii) \quad \overline{AB} = \bar{A} \bar{B}$$

$$(iii) \quad \overline{A+B} = \bar{A} + \bar{B}$$

$$(iv) \quad \overline{\lambda A} = \bar{\lambda} \bar{A}$$

Transpose of Conjugate of a Matrix (A^θ or A^*)

$$A^\theta = (\bar{A})^T$$

Example : if $A = \begin{bmatrix} 2 & 1+i \\ 1-i & 5 \end{bmatrix}$ then find A^θ

$$\bar{A} = \begin{bmatrix} 2 & 2-i \\ 1+i & 5 \end{bmatrix}$$

$$(\bar{A})^T = \begin{bmatrix} 2 & 1+i \\ 2-i & 5 \end{bmatrix}$$

$$A^\theta = \begin{bmatrix} 2 & 1+i \\ 2-i & 5 \end{bmatrix}$$

Note :

$$(i) (A^\theta)^\theta = A$$

$$(ii) (AB)^\theta = B^\theta A^\theta$$

$$(iii) (A+B)^\theta = A^\theta + B^\theta$$

$$(iv) (\lambda A)^\theta = \bar{\lambda} A^\theta$$

Types of Complex Matrices

(i) Hermitian Matrix

- A Square matrix A is said to be Hermitian if

$$A^{\theta} = A$$

- In Hermitian matrix $A = [a_{ij}]$

$$a_{ij} = \overline{a_{ji}} \quad \text{for all } i, j$$

- The diagonal elements of a Hermitian matrix are always real

Example : $A = \begin{bmatrix} 2 & 4-i \\ 4+i & 5 \end{bmatrix}$

$$\bar{A} = \begin{bmatrix} 2 & 4+i \\ 4-i & 5 \end{bmatrix}$$

$$(\bar{A})' = \begin{bmatrix} 2 & 4-i \\ 4+i & 5 \end{bmatrix}$$

$$A^{\theta} = A$$

Hence, A is a Hermitian matrix

(ii) Skew-Hermitian Matrix

- A Square matrix A is said to be skew-Hermitian if

$$A^{\theta} = -A$$

- In skew Hermitian matrix $A = [a_{ij}]$

$$a_{ij} = -\bar{a}_{ji} \quad \text{for all } i, j$$

- The diagonal elements of a skew-Hermitian matrix are either zero or purely imaginary

Example : $A = \begin{bmatrix} 0 & 1+i \\ -1+i & 0 \end{bmatrix}$

$$\bar{A} = \begin{bmatrix} 0 & 1-i \\ -1-i & 0 \end{bmatrix}$$

$$(\bar{A})' = \begin{bmatrix} 0 & -1-i \\ 1-i & 0 \end{bmatrix}$$

$$A^{\theta} = -\begin{bmatrix} 0 & 1+i \\ -1+i & 0 \end{bmatrix}$$

$$A^{\theta} = -A$$

Hence, A is Skew-Hermitian

Q.1 If A is a Hermitian matrix, then show that iA is a skew-Hermitian.

AKTU 2023

A is a Hermitian Matrix

$$A^{\theta} = A \quad \text{--- ①}$$

$$(iA)^{\theta} = \bar{i} A^{\theta} \quad \left\{ \because (\lambda A)^{\theta} = \bar{\lambda} A^{\theta} \right.$$

$$(iA)^{\theta} = -i A^{\theta}$$

$$(iA)^{\theta} = -i A$$

Hence, iA is a skew Hermitian matrix

Q.2 If A is a skew-Hermitian matrix, then show that iA is Hermitian.

Given

$$A^{\theta} = -A$$

$$(iA)^{\theta} = \bar{i} A^{\theta} \quad \left\{ (\lambda A)^{\theta} = \bar{\lambda} A^{\theta} \right.$$

$$(iA)^{\theta} = -i A^{\theta}$$

$$(iA)^{\theta} = (-i)(-A)$$

$$(iA)^{\theta} = iA$$

Hence, iA is a Hermitian matrix

Q.3 Show that the matrix A is Hermitian and iA is skew-Hermitian where $A = \begin{bmatrix} 2 & 3-4i \\ 3+4i & 2 \end{bmatrix}$

To show that A is Hermitian

$$A = \begin{bmatrix} 2 & 3-4i \\ 3+4i & 2 \end{bmatrix}$$

$$\bar{A} = \begin{bmatrix} 2 & 3+4i \\ 3-4i & 2 \end{bmatrix}$$

$$A^{\theta} = \begin{bmatrix} 2 & 3-4i \\ 3+4i & 2 \end{bmatrix}$$

$$\boxed{A^{\theta} = A}$$

Hence, A is a Hermitian matrix

To show that iA is skew-Hermitian

$$iA = i \begin{bmatrix} 2 & 3-4i \\ 3+4i & 2 \end{bmatrix}$$

$$iA = \begin{bmatrix} 2i & 3i-4i^2 \\ 3i+4i^2 & 2i \end{bmatrix}$$

$$iA = \begin{bmatrix} 2i & 4+3i \\ -4+3i & 2i \end{bmatrix} \quad \text{--- ①}$$

$$\overline{iA} = \begin{bmatrix} -2i & 4-3i \\ -4-3i & -2i \end{bmatrix}$$

$$(\overline{iA})' = \begin{bmatrix} -2i & -4-3i \\ 4-3i & -2i \end{bmatrix}$$

$$(iA)^{\theta} = - \begin{bmatrix} 2i & 4+3i \\ -4+3i & 2i \end{bmatrix}$$

$$(iA)^{\theta} = -(iA)$$

Hence

iA is a skew

Hermitian matrix

Every Square matrix can be uniquely expressed as the sum of a Hermitian matrix and a Skew-Hermitian matrix

Let A be any Square matrix

$$A = \frac{1}{2}A + \frac{1}{2}A$$

$$A = \frac{1}{2}A + \frac{1}{2}A^{\theta} + \frac{1}{2}A - \frac{1}{2}A^{\theta}$$

$$A = \frac{1}{2}(A + A^{\theta}) + \frac{1}{2}(A - A^{\theta})$$

$\uparrow$
 P

$\uparrow$
 Q

$$A = P + Q$$

Hermitian
Matrix

Skew-Hermitian
Matrix

$$P = \frac{1}{2}(A + A^{\theta})$$

$$Q = \frac{1}{2}(A - A^{\theta})$$

Q.4 Express the matrix $A = \begin{bmatrix} i & 2-3i & 4+5i \\ 6+i & 0 & 4-5i \\ -i & 2-i & 2+i \end{bmatrix}$, as a sum of Hermitian and Skew-Hermitian matrix

$$A = \begin{bmatrix} i & 2-3i & 4+5i \\ 6+i & 0 & 4-5i \\ -i & 2-i & 2+i \end{bmatrix} \quad A + A^{\theta} = \begin{bmatrix} i & 2-3i & 4+5i \\ 6+i & 0 & 4-5i \\ -i & 2-i & 2+i \end{bmatrix} + \begin{bmatrix} -i & 6-i & i \\ 2+3i & 0 & 2+i \\ 4-5i & 4+5i & 2-i \end{bmatrix}$$

$$\bar{A} = \begin{bmatrix} -i & 2+3i & 4-5i \\ 6-i & 0 & 4+5i \\ i & 2+i & 2-i \end{bmatrix} \quad A + A^{\theta} = \begin{bmatrix} 0 & 8-4i & 4+6i \\ 8+4i & 0 & 6-4i \\ 4-6i & 6+4i & 4 \end{bmatrix}$$

$$A^{\theta} = \begin{bmatrix} -i & 6-i & i \\ 2+3i & 0 & 2+i \\ 4-5i & 4+5i & 2-i \end{bmatrix} \quad P = \frac{1}{2}(A + A^{\theta}) = \frac{1}{2} \begin{bmatrix} 0 & 8-4i & 4+6i \\ 8+4i & 0 & 6-4i \\ 4-6i & 6+4i & 4 \end{bmatrix}$$

$$P = \begin{bmatrix} 0 & 4-2i & 2+3i \\ 4+2i & 0 & 3-2i \\ 2-3i & 3+2i & 4 \end{bmatrix}$$

P is a Hermitian matrix

$$A - A^{\theta} = \begin{bmatrix} i & 2-3i & 4+5i \\ 6+i & 0 & 4-5i \\ -i & 2-i & 2+i \end{bmatrix} - \begin{bmatrix} -i & 6-i & i \\ 2+3i & 0 & 2+i \\ 4-5i & 4+5i & 2-i \end{bmatrix}$$

$$A - A^{\theta} = \begin{bmatrix} 2i & -4-2i & 4+4i \\ 4-2i & 0 & 2-6i \\ -4+4i & -2-6i & 2i \end{bmatrix}$$

$$Q = \frac{1}{2}(A - A^{\theta})$$

$$Q = \begin{bmatrix} i & -2-i & 2+2i \\ 2-i & 0 & 1-3i \\ -2+2i & -1-3i & 2i \end{bmatrix}$$

Q is a skew-Hermitian

$$A = P + Q$$

Hence,

A can be expressed as a sum of Hermitian and skew Hermitian

3. Unitary Matrix

A Square matrix A is said to be unitary if

$$A^{\theta} A = A A^{\theta} = I$$

$$A A^{\theta} = I$$

$$A^{-1} A A^{\theta} = A^{-1} I$$

$$I A^{\theta} = A^{-1}$$

$$A^{\theta} = A^{-1}$$

$$A^{-1} = A^{\theta}$$

Q.5 Show that the following matrix is unitary:

$$\frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix}$$

AKTU 2021

$$A = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix}$$

$$\bar{A} = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1-i \\ 1+i & -1 \end{bmatrix}$$

$$A^T = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix}$$

$$AA^T = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix} \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix}$$

$$AA^T = \frac{1}{3} \begin{bmatrix} 1+1-i^2 & 1+i-1-i \\ 1-i-1+i & 1-i^2+1 \end{bmatrix}$$

$$= \frac{1}{3} \begin{bmatrix} 1+1+1 & 0 \\ 0 & 1+1+1 \end{bmatrix}$$

$$AA^T = \frac{1}{3} \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} = \frac{1}{3} \times 3 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$AA^T = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$AA^T = I$$

Hence,

A is unitary matrix

Q.6 Show that the matrix $\begin{bmatrix} \alpha + i\gamma & -\beta + i\delta \\ \beta + i\delta & \alpha - i\gamma \end{bmatrix}$ is unitary if $\alpha^2 + \beta^2 + \gamma^2 + \delta^2 = 1$

AKTU 2018

$$A = \begin{bmatrix} \alpha + i\gamma & -\beta + i\delta \\ \beta + i\delta & \alpha - i\gamma \end{bmatrix}$$

$$\bar{A} = \begin{bmatrix} \alpha - i\gamma & -\beta - i\delta \\ \beta - i\delta & \alpha + i\gamma \end{bmatrix}$$

$$A^{\theta} = \begin{bmatrix} \alpha - i\gamma & \beta - i\delta \\ -\beta - i\delta & \alpha + i\gamma \end{bmatrix}$$

$$AA^{\theta} = \begin{bmatrix} \alpha + i\gamma & -\beta + i\delta \\ \beta + i\delta & \alpha - i\gamma \end{bmatrix} \begin{bmatrix} \alpha - i\gamma & \beta - i\delta \\ -\beta - i\delta & \alpha + i\gamma \end{bmatrix}$$

$$AA^{\theta} = \begin{bmatrix} \alpha^2 + \gamma^2 + \beta^2 + \delta^2 & (\alpha + i\gamma)(\beta - i\delta) - (\beta - i\delta)(\alpha + i\gamma) \\ (\beta + i\delta)(\alpha - i\gamma) - (\beta + i\delta)(\alpha - i\gamma) & \beta^2 + \delta^2 + \alpha^2 + \gamma^2 \end{bmatrix}$$

$$AA^{\theta} = \begin{bmatrix} \alpha^2 + \beta^2 + \gamma^2 + \delta^2 & 0 \\ 0 & \alpha^2 + \beta^2 + \gamma^2 + \delta^2 \end{bmatrix}$$

$$\text{If } \alpha^2 + \beta^2 + \gamma^2 + \delta^2 = 1$$

$$AA^{\theta} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

Hence, A is unitary if $\alpha^2 + \beta^2 + \gamma^2 + \delta^2 = 1$

Complex Cube Root of Unity (ω)

$$\omega = (1)^{1/3}$$

$$\omega^3 = 1$$

$$\omega^3 - 1 = 0$$

$$\omega^3 - 1^3 = 0$$

$$(\omega - 1)(\omega^2 + \omega + 1) = 0$$

$$\omega = 1$$

$$\omega^2 + \omega + 1 = 0$$

$$\omega = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\omega = \frac{-1 \pm \sqrt{1-4}}{2}$$

$$\omega = \frac{-1 \pm i\sqrt{3}}{2}$$

Hence

$$\omega = 1$$

$$\omega = \frac{-1 + i\sqrt{3}}{2} = \omega$$

$$\omega = \frac{-1 - i\sqrt{3}}{2} = \omega^2$$

Note

$$\overline{\omega} = \omega^2$$

$$\overline{\omega^2} = \omega$$

$$\omega^3 = 1$$

$$1 + \omega + \omega^2 = 0$$

Q.7 Show that

$$A = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{bmatrix}$$

is a unitary matrix, where ω is complex cube root of unity

AKTU 2017

$$A = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{bmatrix}$$

$$\bar{A} = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1 & 1 \\ 1 & \omega^2 & \omega \\ 1 & \omega & \omega^2 \end{bmatrix}$$

$$A^{\theta} = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1 & 1 \\ 1 & \omega^2 & \omega \\ 1 & \omega & \omega^2 \end{bmatrix}$$

$$AA^{\theta} = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{bmatrix} \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1 & 1 \\ 1 & \omega^2 & \omega \\ 1 & \omega & \omega^2 \end{bmatrix}$$

$$AA^{\theta} = \frac{1}{3} \begin{bmatrix} 1+1+1 & 1+\omega^2+\omega & 1+\omega+\omega^2 \\ 1+\omega+\omega^2 & 1+\omega^3+\omega^3 & 1+\omega^2+\omega^4 \\ 1+\omega^2+\omega & 1+\omega^4+\omega^2 & 1+\omega^3+\omega^3 \end{bmatrix}$$

$$AA^{\theta} = \frac{1}{3} \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} = \frac{3}{3} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$AA^{\theta} = I$$

Hence, A is unitary matrix

Matrix : DPP-9

Topic : Complex Matrices

Q.1 Show that the matrix A is Hermitian and iA is skew- Hermitian where A is
$$\begin{bmatrix} 2 & 3+2i & -4 \\ 3-2i & 5 & 6i \\ -4 & -6i & 3 \end{bmatrix}$$
 AKTU 2014

Q.2 If $A = \begin{bmatrix} 2+i & 3 & -1+3i \\ -5 & i & 4-2i \end{bmatrix}$, verify that A^*A is a Hermitian matrix where A^* is the conjugate transpose of A.

Q.3 Express the Hermitian matrix $A = \begin{bmatrix} 1 & -i & 1+i \\ i & 0 & 2-3i \\ 1-i & 2+3i & 2 \end{bmatrix}$ as $P + iQ$, where P is a real symmetric and Q is real skew-symmetric.

Q.4 Show that the matrix $A = \frac{1}{2} \begin{bmatrix} 1+i & -1+i \\ 1+i & 1-i \end{bmatrix}$ is unitary. Also find A^{-1}

ENGINEERING MATHEMATICS - I

UNIT -1 : MATRICES

Lecture -10

Today's target

- ✓ Characteristics Equations
- ✓ Eigen Values
- ✓ Properties of Eigen Values
- PYQs
- DPP



By Gulshan sir

For 2 X 2 order matrix A

Method - 1

$$|A - \lambda I| = 0$$

Expand the determinant

Method - 2

$$|A - \lambda I| = 0$$

$$\lambda^2 - S\lambda + |A| = 0$$

Where

 S = Trace of the matrix $|A|$ = Determinant of the matrix

For 3 X 3 order matrix A

Method - 1

$$|A - \lambda I| = 0$$

Expand the determinant

Method - 2

$$|A - \lambda I| = 0$$

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - |A| = 0$$

Where

 S_1 = Trace of the matrix

$$S_2 = M_{11} + M_{22} + M_{33}$$

 $|A|$ = Determinant of the matrix

Eigen Values OR Latent root OR Characteristic roots : Eigen values are the roots of Characteristic Equation

Q.1 Find the eigen values of the matrix $A = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$

Method-1

$$A = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$$

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 5-\lambda & 4 \\ 1 & 2-\lambda \end{vmatrix} = 0$$

Expand

$$(5-\lambda)(2-\lambda) - 4 = 0$$

$$10 - 5\lambda - 2\lambda + \lambda^2 - 4 = 0$$

$$\lambda^2 - 7\lambda + 6 = 0$$

Eigen values

$$\lambda = 6, 1$$

Method-2

$$A = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$$

$$|A - \lambda I| = 0$$

$$\lambda^2 - S\lambda + |A| = 0$$

$$S = \text{Trace of matrix} \\ = 7$$

$$|A| = \begin{vmatrix} 5 & 4 \\ 1 & 2 \end{vmatrix}$$

$$= 10 - 4$$

$$= 6$$

$$\lambda^2 - 7\lambda + 6 = 0$$

Eigen values

$$\lambda = 6, 1$$

$$ax^2 + bx + c = 0$$

Q. 2 Find the eigen values of the matrix $\begin{bmatrix} -5 & 2 \\ 2 & -2 \end{bmatrix}$

$$A = \begin{bmatrix} -5 & 2 \\ 2 & -2 \end{bmatrix}$$

$$|A - \lambda I| = 0$$

$$\lambda^2 - S\lambda + |A| = 0$$

$$S = \text{Trace of Matrix}$$
$$= -5 - 2$$

$$= -7$$

$$|A| = \begin{vmatrix} -5 & 2 \\ 2 & -2 \end{vmatrix}$$

$$|A| = (-5)(-2) - 4$$

$$= 10 - 4$$

$$= 6$$

$$\lambda^2 - (-7)\lambda + 6 = 0$$

$$\lambda^2 + 7\lambda + 6 = 0$$

Eigen values

$$\lambda = -1, -6$$

$$ax^3 + bx^2 + cx + d = 0$$

Q.3 Find the eigen values of the matrix

$$\begin{bmatrix} 1 & 0 & 4 \\ 0 & 2 & 0 \\ 3 & 1 & -3 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 0 & 4 \\ 0 & 2 & 0 \\ 3 & 1 & -3 \end{bmatrix}$$

$$|A - \lambda I| = 0$$

$$\lambda^3 - S_1 \lambda^2 + S_2 \lambda - |A| = 0 \quad \text{--- ①}$$

S_1 = Trace of matrix

$$S_1 = 1 + 2 - 3$$

$$S_1 = 0$$

$$S_2 = M_{11} + M_{22} + M_{33}$$

$$S_2 = \begin{vmatrix} 2 & 0 \\ 1 & -3 \end{vmatrix} + \begin{vmatrix} 1 & 4 \\ 3 & -3 \end{vmatrix} + \begin{vmatrix} 1 & 0 \\ 0 & 2 \end{vmatrix}$$

$$S_2 = -6 - 0 + (-3 - 12) + (2 - 0)$$

$$S_2 = -6 - 15 + 2$$

$$S_2 = -19$$

$$|A| = \begin{vmatrix} 1 & 0 & 4 \\ 0 & 2 & 0 \\ 3 & 1 & -3 \end{vmatrix}$$

Expand along R_1

$$|A| = 1(-6 - 0) - 0 + 4(0 - 6)$$

$$|A| = -6 - 24$$

$$|A| = -30$$

$$\lambda^3 - 19\lambda + 30 = 0$$

Eigen values

$$\lambda = 3, -5, 2$$

Some important properties of eigen values

1. The sum of eigen values of a matrix is equal to trace of the matrix.

2. Determinant of matrix is equal to the product of eigen values.

3. If $\lambda_1, \lambda_2, \lambda_3, \dots, \lambda_n$ are the eigen values of A, then the eigen values of

(i) kA are $k\lambda_1, k\lambda_2, k\lambda_3, \dots, k\lambda_n$

(ii) A^m are $\lambda_1^m, \lambda_2^m, \lambda_3^m, \dots, \lambda_n^m$

(iii) A^{-1} are $\frac{1}{\lambda_1}, \frac{1}{\lambda_2}, \frac{1}{\lambda_3}, \dots, \frac{1}{\lambda_n}$

4. Any square matrix A and its transpose A' have the same Eigen values.

5. The Eigen values of a Hermitian matrix are all real

6. The Eigen values of a Skew-Hermitian matrix is either zero or purely an imaginary number

7. The eigen values of a diagonal matrix are just the diagonal elements.

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -4 & 0 \\ 0 & 0 & 6 \end{bmatrix}$$

$$\lambda = 2, -4, 6$$

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\lambda = 1, 1, 1$$

8. The eigen values of a triangular matrix are just the diagonal elements of the matrix.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & -6 \\ 0 & 0 & 5 \end{bmatrix}$$

$$\lambda = 1, 4, 5$$

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 2 & 0 \\ 3 & 3 & 5 \end{bmatrix}$$

$$\lambda = 1, 2, 5$$

Q.4 Find the product and sum of the Eigen values for $A = \begin{bmatrix} 8 & -4 \\ 2 & 2 \end{bmatrix}$

$$A = \begin{bmatrix} 8 & -4 \\ 2 & 2 \end{bmatrix}$$

Product of Eigen values = $|A|$

$$= \begin{vmatrix} 8 & -4 \\ 2 & 2 \end{vmatrix}$$

$$= 16 + 8$$

$$= 24$$

Sum of Eigen values = Trace of A

$$= 8 + 2$$

$$= 10$$

Q.5 If 2 and 3 are two eigen values of matrix A and determinant of A is 24. Find the third eigen value.

Let third Eigen value = λ

Product of Eigen values = $|A|$

$$2 \times 3 \times \lambda = 24$$

$$\lambda = \frac{24}{6}$$

$$\lambda = 4$$

Q.6 Find the value of x , the Eigen values of the Given matrix A are real

$$A = \begin{bmatrix} 10 & 5+i & 4 \\ x & 20 & 2 \\ 4 & 2 & -10 \end{bmatrix}$$

AKTU 2017

We know that

Eigen values of a Hermitian matrix are real

Hence, A is a Hermitian matrix

$\Rightarrow x$ is the conjugate of $5+i$

$$x = 5-i$$

Q.7 Find the eigen values of A^3 where $A = \begin{bmatrix} 3 & 2 \\ 1 & 2 \end{bmatrix}$

$$A = \begin{bmatrix} 3 & 2 \\ 1 & 2 \end{bmatrix}$$

$$|A - \lambda I| = 0$$

$$\lambda^2 - S\lambda + |A| = 0$$

$S = \text{Trace of matrix}$

$$S = 5$$

$$|A| = \begin{vmatrix} 3 & 2 \\ 1 & 2 \end{vmatrix}$$

$$|A| = 6 - 2 = 4$$

$$\lambda^2 - 5\lambda + 4 = 0$$

$$\lambda = 4, 1$$

Eigen values of $A = 1, 4$

$$A^3 = \begin{bmatrix} 1^3 & 4^3 \\ 1 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 64 \\ 1 & 4 \end{bmatrix}$$

Two eigen values of the matrix $\begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$ are equal to 1 each. Find the eigen values of A^{-1}

$$A = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$$

$$\lambda_1 = 1$$

$$\lambda_2 = 1$$

$$\text{Let } \lambda_3 = \lambda$$

We know that

$$\lambda_1 + \lambda_2 + \lambda_3 = \text{Trace of matrix}$$

$$1 + 1 + \lambda = 2 + 3 + 2$$

$$2 + \lambda = 7$$

$$\lambda = 5$$

Hence

$$\lambda_1 = 1$$

$$\lambda_2 = 1$$

$$\lambda_3 = 5$$

Eigen values of $A = 1, 1, 5$

$$A^{-1} = \frac{1}{1}, \frac{1}{1}, \frac{1}{5}$$

$$= 1, 1, \frac{1}{5}$$

Q.9 If the matrix $A = \begin{bmatrix} -1 & 2 & 3 \\ 0 & 3 & 5 \\ 0 & 0 & -2 \end{bmatrix}$ then find the eigen values of $A^3 + 5A + 8I$

AKTU 2022

$$A = \begin{bmatrix} -1 & 2 & 3 \\ 0 & 3 & 5 \\ 0 & 0 & -2 \end{bmatrix}$$

Since, A is a triangular matrix

Eigen values of A

$$\lambda = -1, 3, -2$$

When $\lambda = -1$

$$\begin{aligned} A^3 + 5A + 8I &= (-1)^3 + 5(-1) + 8(1) \\ &= -1 - 5 + 8 \\ &= 2 \end{aligned}$$

When $\lambda = 3$

$$\begin{aligned} A^3 + 5A + 8I &= (3)^3 + 5 \times 3 + 8 \times 1 \\ &= 27 + 15 + 8 \\ &= 50 \end{aligned}$$

When $\lambda = -2$

$$A^3 + 5A + 8I = (-2)^3 + 5(-2) + 8(1)$$

$$\begin{aligned} A^3 + 5A + 8I &= -18 + 8 \\ &= -10 \end{aligned}$$

Hence

Eigen values are
 $= 2, 50, -10$

Q.10 If the Eigen values of matrix A are 1, 1, 1 then find the Eigen values of $A^2 + 2A + 3I$

AKTU 2019

Eigen values of A

$$\lambda = 1, 1, 1$$

When $\lambda = 1$

$$\begin{aligned} A^2 + 2A + 3I &= (1)^2 + 2(1) + 3(1) \\ &= 1 + 2 + 3 \\ &= 6 \end{aligned}$$

Hence

Eigen values of $A^2 + 2A + 3I$

$$= 6, 6, 6$$

Topic : Eigen Values

Q.1 Find the eigen values of the matrix $\begin{bmatrix} 2 & -3 & 1 \\ 3 & 1 & 3 \\ -5 & 2 & -4 \end{bmatrix}$

Ans 0, 1, -2

Q.2 Find the eigen values of the matrix $\begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$

Ans 1, 2, 3

Q.3 The matrix A is defined as $\begin{bmatrix} 1 & 2 & -3 \\ 0 & 2 & 2 \\ 0 & 0 & -2 \end{bmatrix}$ Find the eigen values of $3A^3 + 5A^2 - 6A + 2I$

Ans 4, 110, 10

Q.4 The matrix $A = \begin{bmatrix} 1 & 0 \\ 2 & 4 \end{bmatrix}$ is given. Find the eigen values of $4A^{-1} + 3A + 2I$

Ans 6, 15

ENGINEERING MATHEMATICS - I

UNIT -1 : MATRICES

Lecture - 11

Today's target

- Eigen Values and Eigen Vectors
- PYQs
- DPP



By Gulshan sir

Eigen vectors corresponding to an Eigen value is not unique(i.e. Eigen vectors are infinite)

Q.1 Find the eigen values and eigen vectors the matrix $\begin{bmatrix} -5 & 2 \\ 2 & -2 \end{bmatrix}$

$$A = \begin{bmatrix} -5 & 2 \\ 2 & -2 \end{bmatrix}$$

$$|A - \lambda I| = 0$$

$$\lambda^2 - S\lambda + |A| = 0$$

S = Trace of matrix

$$S = -5 - 2$$

$$S = -7$$

$$|A| = 10 - 4$$

$$|A| = 6$$

$$\lambda^2 + 7\lambda + 6 = 0$$

$$\lambda = -1, -6$$

Non-zero eigen vectors corresponding to eigen values

$$(A - \lambda I)X = 0$$

$$\begin{bmatrix} -5-\lambda & 2 \\ 2 & -2-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

①

When $\lambda = -1$

$$\textcircled{2} \begin{bmatrix} -4 & 2 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$R_2 \rightarrow 2R_2 + R_1$$

$$\begin{bmatrix} -4 & 2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$-4x_1 + 2x_2 = 0$$

Let $x_1 = k_1$

$$-4k_1 + 2x_2 = 0$$

$$2x_2 = 4k_1$$

$$x_2 = 2k_1$$

Eigen vector

$$X_1 = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$X_1 = \begin{bmatrix} k_1 \\ 2k_1 \end{bmatrix}$$

$$X_1 = k_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$k_1 \in \mathbb{R}$$

When $\lambda = -6$

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$R_2 \rightarrow R_2 + (-2R_1)$$

$$\begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$x_1 + 2x_2 = 0$$

Let $x_2 = k_2$

$$x_1 = -2k_2$$

Eigen vector

$$X_2 = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$= \begin{bmatrix} -2k_2 \\ k_2 \end{bmatrix}$$

$$X_2 = k_2 \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

Q.2 Find the Eigen values and corresponding Eigen Vectors of the matrix $A = \begin{bmatrix} 2 & 1 & 1 \\ 2 & 3 & 4 \\ -1 & -1 & -2 \end{bmatrix}$

AKTU 2023

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 2 & 3 & 4 \\ -1 & -1 & -2 \end{bmatrix}$$

$$|A - \lambda I| = 0$$

$$\lambda^3 - S_1 \lambda^2 + S_2 \lambda - |A| = 0$$

$$S_1 = \text{Trace of matrix} \\ = 2 + 3 - 1$$

$$S_1 = 3$$

$$S_2 = M_{11} + M_{22} + M_{33}$$

$$S_2 = \begin{vmatrix} 3 & 4 \\ -1 & -2 \end{vmatrix} + \begin{vmatrix} 2 & 1 \\ -1 & -2 \end{vmatrix} + \begin{vmatrix} 2 & 1 \\ 2 & 3 \end{vmatrix}$$

$$S_2 = -6 + 4 + (-4 + 1) + (6 - 2)$$

$$S_2 = -2 - 3 + 4$$

$$S_2 = -1$$

$$|A| = \begin{vmatrix} 2 & 1 & 1 \\ 2 & 3 & 4 \\ -1 & -1 & -2 \end{vmatrix}$$

$$|A| = 2(-6 + 4) - 1(-4 + 4) + 1(-2 + 3)$$

$$|A| = -4 + 1$$

$$|A| = -3$$

$$\lambda^3 - 3\lambda^2 - \lambda + 3 = 0$$

$$\lambda = 3, -1, 1$$

Eigen vectors corresponding
to eigen values

$$(A - \lambda I)X = 0$$

$$\begin{bmatrix} 2-\lambda & 1 & 1 \\ 2 & 3-\lambda & 4 \\ -1 & -1 & -2-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

When $\lambda = 3$

$$\begin{bmatrix} -1 & 1 & 1 \\ 2 & 0 & 4 \\ -1 & -1 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \text{--- (1)}$$

$$R_2 \rightarrow R_2 + 2R_1$$

$$R_3 \rightarrow R_3 - R_1$$

$$\begin{bmatrix} -1 & 1 & 1 \\ 0 & 2 & 6 \\ 0 & -2 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + R_2$$

$$\begin{bmatrix} -1 & 1 & 1 \\ 0 & 2 & 6 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$-x_1 + x_2 + x_3 = 0 \quad \text{--- (2)}$$

$$2x_2 + 6x_3 = 0 \quad \text{--- (3)}$$

Let $x_3 = k_1$

$$2x_2 + 6k_1 = 0$$

$$2x_2 = -6k_1$$

$$x_2 = -3k_1$$

From (2)

$$-x_1 - 3k_1 + k_1 = 0$$

$$-x_1 - 2k_1 = 0$$

$$x_1 = -2k_1$$

Eigen Vectors

$$X_1 = \begin{bmatrix} n_1 \\ n_2 \\ n_3 \end{bmatrix}$$

$$X_1 = \begin{bmatrix} -2K_1 \\ -3K_1 \\ K_1 \end{bmatrix}$$

$$X_1 = K_1 \begin{bmatrix} -2 \\ -3 \\ 1 \end{bmatrix}$$

$$K_1 \in \mathbb{R}$$

When $\lambda = -1$

$$\begin{bmatrix} 3 & 1 & 1 \\ 2 & 4 & 4 \\ -1 & -1 & -1 \end{bmatrix} \begin{bmatrix} n_1 \\ n_2 \\ n_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$R_1 \leftrightarrow R_3$$

$$\begin{bmatrix} -1 & -1 & -1 \\ \textcircled{2} & 4 & 4 \\ \textcircled{3} & 1 & 1 \end{bmatrix} \begin{bmatrix} n_1 \\ n_2 \\ n_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$R_2 \rightarrow R_2 + 2R_1 \quad R_3 \rightarrow R_3 + 3R_1$$

$$\begin{bmatrix} -1 & -1 & -1 \\ 0 & 2 & 2 \\ 0 & \textcircled{-2} & -2 \end{bmatrix} \begin{bmatrix} n_1 \\ n_2 \\ n_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2$$

$$\begin{bmatrix} -1 & -1 & -1 \\ 0 & 2 & 2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} n_1 \\ n_2 \\ n_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$-n_1 - n_2 - n_3 = 0 \quad \text{--- (3)}$$

$$2n_2 + 2n_3 = 0 \quad \text{--- (4)}$$

$$\text{Let } n_3 = K_2$$

$$n_2 = -K_2$$

$$-n_1 + K_2 - K_2 = 0 \quad n_1 = 0$$

Eigen Vector

$$X_2 = \begin{bmatrix} n_1 \\ n_2 \\ n_3 \end{bmatrix}$$

$$X_2 = \begin{bmatrix} 0 \\ -k_2 \\ k_2 \end{bmatrix}$$

$$X_2 = k_2 \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$$

$$k_2 \in \mathbb{R}$$

When $\lambda = 1$

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 4 \\ -1 & -1 & -3 \end{bmatrix} \begin{bmatrix} n_1 \\ n_2 \\ n_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$R_2 \rightarrow R_2 + (-2R_1)$$

$$R_3 \rightarrow R_3 + R_1$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 2 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} n_1 \\ n_2 \\ n_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$n_1 + n_2 + n_3 = 0 \quad \text{--- (5)}$$

$$2n_3 = 0 \quad \text{--- (6)}$$

$$-2n_3 = 0 \quad \text{--- (7)}$$

$$n_3 = 0$$

From (5)

$$n_1 + n_2 + 0 = 0$$

$$n_1 + n_2 = 0$$

$$n_1 = -n_2$$

$$\frac{n_1}{1} = \frac{n_2}{-1} = k_3$$

$$n_1 = k_3$$

$$n_2 = -k_3$$

Eigen vector

$$X_3 = \begin{bmatrix} n_1 \\ n_2 \\ n_3 \end{bmatrix}$$

$$X_3 = \begin{bmatrix} k_3 \\ -k_3 \\ 0 \end{bmatrix}$$

$$X_3 = k_3 \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

Q.3 Find the Eigen values and corresponding Eigen Vectors of the matrix $A = \begin{bmatrix} 1 & 2 & 2 \\ 0 & 2 & 1 \\ -1 & 2 & 2 \end{bmatrix}$ AKTU 2024

$$A = \begin{bmatrix} 1 & 2 & 2 \\ 0 & 2 & 1 \\ -1 & 2 & 2 \end{bmatrix}$$

$$|A - \lambda I| = 0$$

$$\lambda^3 - S_1 \lambda^2 + S_2 \lambda - |A| = 0$$

$$S_1 = \text{Trace of matrix} \\ = 1 + 2 + 2$$

$$S_1 = 5$$

$$S_2 = M_{11} + M_{22} + M_{33}$$

$$S_2 = 2 + 4 + 2$$

$$S_2 = 8$$

$$|A| = \begin{vmatrix} 1 & 2 & 2 \\ 0 & 2 & 1 \\ -1 & 2 & 2 \end{vmatrix}$$

$$|A| = 1(4-2) - 2(0+1) + 2(0+2)$$

$$|A| = 2 - 2 + 4$$

$$|A| = 4$$

$$A = \begin{bmatrix} 1 & 2 & 2 \\ 0 & 2 & 1 \\ -1 & 2 & 2 \end{bmatrix}$$

$$\lambda^3 - 5\lambda^2 + 8\lambda - 4 = 0$$

$$\lambda = 1, 2, 2$$

Non-zero eigen vectors
corresponding eigen
values

$$(A - \lambda I)X = 0$$

$$\begin{bmatrix} 1-\lambda & 2 & 2 \\ 0 & 2-\lambda & 1 \\ -1 & 2 & 2-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

①

When $\lambda = 1$

$$\begin{bmatrix} 0 & 2 & 2 \\ 0 & 1 & 1 \\ -1 & 2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$R_1 \leftrightarrow R_3$$

$$\begin{bmatrix} -1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 2 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + (-2R_2)$$

$$\begin{bmatrix} -1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$-x_1 + 2x_2 + x_3 = 0 \quad \text{--- (2)}$$

$$x_2 + x_3 = 0 \quad \text{--- (3)}$$

$$\text{Let } x_2 = k_1$$

$$x_3 = -k_1$$

From ②

$$-x_1 + 2k_1 - k_1 = 0$$

$$-x_1 + k_1 = 0$$

$$x_1 = k_1$$

Eigen vector

$$X_1 = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$X_1 = \begin{bmatrix} k_1 \\ k_1 \\ -k_1 \end{bmatrix}$$

$$X_1 = k_1 \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

When $\lambda = 2, 2$

$$\begin{bmatrix} -1 & 2 & 2 \\ 0 & 0 & 1 \\ -1 & 2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_1$$

$$\begin{bmatrix} -1 & 2 & 2 \\ 0 & 0 & 1 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$-x_1 + 2x_2 + 2x_3 = 0 \quad \text{--- (4)}$$

$$x_3 = 0 \quad \text{--- (5)}$$

$$-2x_3 = 0 \quad \text{--- (6)}$$

From (5) and (6)

$$x_3 = 0$$

From (4)

$$-x_1 + 2x_2 + 0 = 0$$

$$-x_1 + 2x_2 = 0$$

$$x_1 = 2x_2$$

$$\frac{x_1}{2} = \frac{x_2}{1} = k_2 \text{ (say)}$$

$$x_1 = 2k_2$$

$$x_2 = k_2$$

$$X_2 = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2k_2 \\ k_2 \\ 0 \end{bmatrix}$$

$$X_2 = k_2 \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

Number of independent vector is less than 3

Q.4 Find all the eigen values and eigen vectors of the matrix $A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 3 & 0 \\ 1 & 0 & 2 \end{bmatrix}$

AKTU 2021

$$A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 3 & 0 \\ 1 & 0 & 2 \end{bmatrix}$$

$$|A - \lambda I| = 0$$

$$\lambda^3 - S_1 \lambda^2 + S_2 \lambda - |A| = 0$$

$$\begin{aligned} S_1 &= \text{Trace of matrix} \\ &= 2 + 3 + 2 \\ &= 7 \end{aligned}$$

$$S_2 = M_{11} + M_{22} + M_{33}$$

$$S_2 = 6 + 3 + 6$$

$$S_2 = 15$$

$$|A| = \begin{vmatrix} 2 & 0 & 1 \\ 0 & 3 & 0 \\ 1 & 0 & 2 \end{vmatrix}$$

$$|A| = 2(6 - 0) + 1(0 - 3)$$

$$|A| = 12 - 3$$

$$|A| = 9$$

$$A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 3 & 0 \\ 1 & 0 & 2 \end{bmatrix}$$

$$\lambda^3 - 7\lambda^2 + 15\lambda - 9 = 0$$

$$\lambda = 1, 3, 3$$

Non-zero eigen vectors
corresponding to eigen values

$$(A - \lambda I)X = 0$$

$$\begin{bmatrix} 2-\lambda & 0 & 1 \\ 0 & 3-\lambda & 0 \\ 1 & 0 & 2-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \text{--- (1)}$$

When $\lambda = 1$

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_1$$

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$x_1 + x_3 = 0 \quad \text{--- (2)}$$

$$2x_2 = 0 \quad \text{--- (3)}$$

From (2)

$$x_2 = 0$$

$$\text{Let } x_1 = k_1$$

$$x_3 = -k_1$$

Eigen vector

$$X_1 = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} k_1 \\ 0 \\ -k_1 \end{bmatrix}$$

$$X_1 = k_1 \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

When $\lambda = 3, 3$

$$\begin{bmatrix} -1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + R_1$$

$$\begin{bmatrix} -1 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$-x_1 + x_3 = 0 \quad \text{--- (4)}$$

$$\text{Let } x_2 = k_2$$

Let $n_1 = k_3$

From (4)

$n_3 = k_3$

$$X = \begin{bmatrix} n_1 \\ n_2 \\ n_3 \end{bmatrix}$$

$$X = \begin{bmatrix} k_3 \\ k_2 \\ k_3 \end{bmatrix}$$

$$X = \begin{bmatrix} 0k_2 + k_3 \\ k_2 + 0k_3 \\ 0k_2 + k_3 \end{bmatrix}$$

$$X = \begin{bmatrix} 0k_2 \\ k_2 \\ 0k_2 \end{bmatrix} + \begin{bmatrix} k_3 \\ 0k_3 \\ k_3 \end{bmatrix}$$

Where

$$X_2 = \begin{bmatrix} 0k_2 \\ k_2 \\ 0k_2 \end{bmatrix}$$

$$X_2 = k_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$X_3 = \begin{bmatrix} k_3 \\ 0k_3 \\ k_3 \end{bmatrix}$$

$$X_3 = k_3 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

Q.5 Find the eigen value of the matrix $\begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix}$ corresponding to the eigen vector $\begin{bmatrix} 51 \\ 51 \end{bmatrix}$

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$$A = \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix}$$

$$X = \begin{bmatrix} 51 \\ 51 \end{bmatrix}$$

We know that

$$(A - \lambda I)X = 0$$

$$\begin{bmatrix} 4-\lambda & 2 \\ 2 & 4-\lambda \end{bmatrix} \begin{bmatrix} 51 \\ 51 \end{bmatrix} = 0$$

$$51(4-\lambda) + 51 \times 2 = 0$$

$$204 - 51\lambda + 102 = 0$$

$$+ 51\lambda = +306$$

$$\lambda = 6$$

Topic : Eigen Values and Eigen Vectors

Q.1 Find the eigen value of the matrix $\begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix}$ corresponding to the eigen vector $\begin{bmatrix} 101 \\ 101 \end{bmatrix}$ **AKTU 2017**

Q.2 Find the eigen values and eigen vectors of the matrix $A = \begin{bmatrix} 3 & 1 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 5 \end{bmatrix}$

Q.3 Find all the eigen values and eigen vectors of the matrix $\begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$

Q.4 Find the eigen values and corresponding eigen vectors of the following matrix $A = \begin{bmatrix} 2 & 1 & 1 \\ 2 & 3 & 2 \\ 3 & 3 & 4 \end{bmatrix}$ **AKTU 2022**

Q.5 Find all the eigen values and eigen vectors of the matrix $A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$

ENGINEERING MATHEMATICS - I

UNIT -1 : MATRICES

Lecture - 12

Today's target

- Cayley-Hamilton Theorem
- PYQs
- DPP



By Gulshan sir

Cayley-Hamilton Theorem

Every square matrix satisfies its own characteristic equation

A

Application of Cayley-Hamilton Theorem

- To find the inverse of the matrix
- To evaluate polynomial and value of polynomial
- To find adjoint of the matrix

$$\lambda^3 - 5\lambda^2 + 4\lambda - 10 = 0$$

$$A^3 - 5A^2 + 4A - 10I = 0$$

$$0 = 0$$

$$A^{-1} = \frac{1}{|A|} \text{adj} \cdot A$$

$$\text{adj} \cdot A = |A| A^{-1}$$

Q.1 Verify Cayley – Hamilton theorem for a matrix $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$ and hence compute A^{-1} .

Also evaluate $A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I$

OR

Q.1 Verify Cayley – Hamilton theorem for a matrix $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$ and hence compute A^{-1} .

Also express the polynomial $B = A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I$ as quadratic polynomial in A and hence find B

OR

Q.1 Verify Cayley – Hamilton theorem for a matrix $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$ and hence compute A^{-1} .

Also find matrix represented by $A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I$ **AKTU 2022**

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$$

$$|A - \lambda I| = 0$$

$$\lambda^3 - S_1 \lambda^2 + S_2 \lambda - |A| = 0$$

$$S_1 = \text{Trace of matrix} \\ = 2 + 1 + 2$$

$$S_1 = 5$$

$$S_2 = M_{11} + M_{22} + M_{33}$$

$$S_2 = \begin{vmatrix} 1 & 0 \\ 1 & 2 \end{vmatrix} + \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} + \begin{vmatrix} 2 & 1 \\ 0 & 1 \end{vmatrix}$$

$$S_2 = 2 + 3 + 2$$

$$S_2 = 7$$

$$|A| = \begin{vmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{vmatrix}$$

$$|A| = \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix}$$

$$|A| = 3$$

$$\lambda^3 - 5\lambda^2 + 7\lambda - 3 = 0$$

By Cayley-Hamilton
Theorem

$$A^3 - 5A^2 + 7A - 3I = 0$$

①

$$A^2 = A \times A$$

$$A^2 = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 4+0+1 & 2+1+1 & 2+0+2 \\ 0+0+0 & 0+1+0 & 0+0+0 \\ 2+0+2 & 1+1+2 & 1+0+4 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 5 & 4 & 4 \\ 0 & 1 & 0 \\ 4 & 4 & 5 \end{bmatrix}$$

$$A^3 = A \times A^2$$

$$A^3 = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} 5 & 4 & 4 \\ 0 & 1 & 0 \\ 4 & 4 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 10+0+4 & 8+1+4 & 8+0+5 \\ 0+0+0 & 0+1+0 & 0+0+0 \\ 5+0+8 & 4+1+8 & 4+0+10 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 14 & 13 & 13 \\ 0 & 1 & 0 \\ 13 & 13 & 14 \end{bmatrix}$$

$$A^3 - 5A^2 + 7A - 3I = 0$$

$$\begin{bmatrix} 14 & 13 & 13 \\ 0 & 1 & 0 \\ 13 & 13 & 14 \end{bmatrix} - 5 \begin{bmatrix} 5 & 4 & 4 \\ 0 & 1 & 0 \\ 4 & 4 & 5 \end{bmatrix} + 7 \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} - 3 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = 0$$

$$\begin{bmatrix} 14 & 13 & 13 \\ 0 & 1 & 0 \\ 13 & 13 & 14 \end{bmatrix} + \begin{bmatrix} -25 & -20 & -20 \\ 0 & -5 & 0 \\ -20 & -20 & -25 \end{bmatrix} + \begin{bmatrix} 14 & 7 & 7 \\ 0 & 7 & 0 \\ 7 & 7 & 14 \end{bmatrix} + \begin{bmatrix} -3 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -3 \end{bmatrix} = 0$$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$LH = RHS$$

Hence

Cayley-Hamilton theorem is

Verified

From ①

$$A^3 - 5A^2 + 7A - 3I = 0$$

Pre-multiply by A^{-1}

$$A^{-1}A^3 - 5A^{-1}A^2 + 7A^{-1}A - 3A^{-1}I = 0$$

$$\textcircled{A^{-1}A}A^2 - 5\textcircled{A^{-1}A}A + 7\textcircled{A^{-1}A} - 3A^{-1}I = 0$$

$$A^2 - 5A + 7I - 3A^{-1} = 0$$

$$3A^{-1} = A^2 - 5A + 7I$$

$$= \begin{bmatrix} 5 & 4 & 4 \\ 0 & 1 & 0 \\ 4 & 4 & 5 \end{bmatrix} - 5 \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} + 7 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$3A^{-1} = \begin{bmatrix} 5 & 4 & 4 \\ 0 & 1 & 0 \\ 4 & 4 & 5 \end{bmatrix} + \begin{bmatrix} -10 & -5 & -5 \\ 0 & -5 & 0 \\ -5 & -5 & -10 \end{bmatrix} + \begin{bmatrix} 7 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 7 \end{bmatrix}$$

$$A^{-1} = \frac{1}{3} \begin{bmatrix} 2 & -1 & -1 \\ 0 & 3 & 0 \\ -1 & -1 & 2 \end{bmatrix}$$

$$\underbrace{A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I}_{= A^2 + A + I}$$

$$= A^5(A^3 - 5A^2 + 7A - 3I) + \underbrace{A^4 - 5A^3 + 8A^2 - 2A + I}_{= \begin{bmatrix} 5 & 4 & 4 \\ 0 & 1 & 0 \\ 4 & 4 & 5 \end{bmatrix} + \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}$$

$$= A^5 \times 0 + A(A^3 - 5A^2) + 8A^2 - 2A + I$$

$$= 0 + A(-7A + 3I) + 8A^2 - 2A + I$$

$$= -7A^2 + 3A + 8A^2 - 2A + I$$

$$= A^2 + A + I$$

$$= \begin{bmatrix} 8 & 5 & 5 \\ 0 & 3 & 0 \\ 5 & 5 & 8 \end{bmatrix}$$

Q.2 Determine A^{-1} , A^{-2} , A^{-3} if $A = \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix}$ by using Cayley – Hamilton theorem

$$A = \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix}$$

$$|A - \lambda I| = 0$$

$$\lambda^3 - S_1 \lambda^2 + S_2 \lambda - |A| = 0$$

$$S_1 = \text{Trace of matrix} \\ = 4 + 3 - 3$$

$$S_1 = 4$$

$$S_2 = M_{11} + M_{22} + M_{33}$$

$$S_2 = \begin{vmatrix} 3 & 2 \\ -4 & -3 \end{vmatrix} + \begin{vmatrix} 4 & 6 \\ -1 & -3 \end{vmatrix} + \begin{vmatrix} 4 & 6 \\ 1 & 3 \end{vmatrix}$$

$$S_2 = (-9 + 8) + (-12 + 6) + (12 - 6)$$

$$S_2 = -1 - 6 + 6$$

$$S_2 = -1$$

$$|A| = \begin{vmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{vmatrix}$$

$$|A| = 4(-9 + 8) - 6(-3 + 2) + 6(-4 + 3)$$

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$$|A| = -4 + 6 - 6$$

$$|A| = -4$$

$$\lambda^3 - 4\lambda^2 - \lambda + 4 = 0$$

By Cayley-Hamilton
theorem

$$A^3 - 4A^2 - A + 4I = 0$$

①

Pre-multiply by A^{-1}

$$A^2 - 4A - I + 4A^{-1} = 0$$

$$4A^{-1} = -A^2 + 4A + I$$

②

$$A^2 = A \times A$$

$$A^2 = \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix} \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} 16+6-6 & 24+18-24 & 24+12-18 \\ 4+3-2 & 6+9-8 & 6+6-6 \\ -4-4+3 & -6-12+12 & -6-8+9 \end{bmatrix}$$

$$= \begin{bmatrix} 16 & 18 & 18 \\ 5 & 7 & 6 \\ -5 & -6 & -5 \end{bmatrix}$$

Put A^2 , A and I in ②

$$4A^{-1} = -\begin{bmatrix} 16 & 18 & 18 \\ 5 & 7 & 6 \\ -5 & -6 & -5 \end{bmatrix} + 4\begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$4A^{-1} = \begin{bmatrix} -16 & -18 & -18 \\ -5 & -7 & -6 \\ 5 & 6 & 5 \end{bmatrix} + \begin{bmatrix} 16 & 24 & 24 \\ 4 & 12 & 8 \\ -4 & -16 & -12 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$4A^{-1} = \begin{bmatrix} 1 & 6 & 6 \\ -1 & 6 & 2 \\ 1 & -10 & -6 \end{bmatrix}$$

$$A^{-1} = \frac{1}{4} \begin{bmatrix} 1 & 6 & 6 \\ -1 & 6 & 2 \\ 1 & -10 & -6 \end{bmatrix}$$

Note

$$A^{-2} = A^{-1} \times A^{-1}$$

$$A^{-3} = A^{-1} \times A^{-2}$$

$$A^3 - 4A^2 - A + 4I = 0 \quad \text{--- ①}$$

Premultiply ① by A^{-2}

$$A - 4I - A^{-1} + 4A^{-2} = 0$$

$$4A^{-2} = -A + 4I + A^{-1}$$

$$16A^{-2} = -4A + 16I + 4A^{-1}$$

$$16A^{-2} = -4 \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix} + 16 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 6 & 6 \\ -1 & 6 & 2 \\ 1 & -10 & -6 \end{bmatrix}$$

$$= \begin{bmatrix} -16 & -24 & -24 \\ -4 & -12 & -8 \\ 4 & 16 & 12 \end{bmatrix} + \begin{bmatrix} 16 & 0 & 0 \\ 0 & 16 & 0 \\ 0 & 0 & 16 \end{bmatrix} + \begin{bmatrix} 1 & 6 & 6 \\ -1 & 6 & 2 \\ 1 & -10 & -6 \end{bmatrix}$$

$$16A^{-2} = \begin{bmatrix} 1 & -18 & -18 \\ -5 & 10 & -6 \\ 5 & 6 & 22 \end{bmatrix}$$

$$A^{-2} = \frac{1}{16} \begin{bmatrix} 1 & -18 & -18 \\ -5 & 10 & -6 \\ 5 & 6 & 22 \end{bmatrix}$$

$$A^3 - 4A^2 - A + 4 = 0 \quad \text{--- (1)}$$

Pre multiply (1) by A^{-3}

$$I - 4A^{-1} - A^{-2} + 4A^{-3} = 0$$

$$4A^{-3} = A^{-2} + 4A^{-1} - I$$

$$64A^{-3} = 16A^{-2} + 16 \times 4A^{-1} - 16I$$

$$64A^{-3} = \begin{bmatrix} 1 & -18 & -18 \\ -5 & 10 & -6 \\ 5 & 6 & 22 \end{bmatrix} + 16 \begin{bmatrix} 1 & 6 & 6 \\ -1 & 6 & 2 \\ 1 & -10 & -6 \end{bmatrix} - 16 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$64A^{-3} = \begin{bmatrix} 1 & -18 & -18 \\ -5 & 10 & -6 \\ 5 & 6 & 22 \end{bmatrix} + \begin{bmatrix} 16 & 96 & 96 \\ -16 & 96 & 32 \\ 16 & -160 & -96 \end{bmatrix} + \begin{bmatrix} -16 & 0 & 0 \\ 0 & -16 & 0 \\ 0 & 0 & -16 \end{bmatrix}$$

$$64A^{-3} = \begin{bmatrix} 1 & 78 & 78 \\ -21 & 90 & 26 \\ 21 & -154 & -90 \end{bmatrix}$$

$$A^{-3} = \frac{1}{64} \begin{bmatrix} 1 & 78 & 78 \\ -21 & 90 & 26 \\ 21 & -154 & -90 \end{bmatrix}$$

Application of matrices to engineering problems

- Graph theory
- Computer Graphics
- Electric circuits with resistors and applied voltages
- A compound mass spring system

Gateway Classes

Topic : Cayley-Hamilton Theorem

Q.1 Verify Cayley-Hamilton theorem for the matrix $A = \begin{bmatrix} 4 & 0 & 1 \\ 0 & 1 & 2 \\ 1 & 0 & 1 \end{bmatrix}$ and find A^{-1}

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$$A^{-1} = \frac{1}{3} \begin{bmatrix} 1 & 0 & -1 \\ 2 & 3 & -8 \\ -1 & 0 & 4 \end{bmatrix}$$

Q.2 Verify Cayley-Hamilton theorem, for the matrix $A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$ and hence compute A^{-1} .

Also evaluate $A^6 - 6A^5 + 9A^4 - 2A^3 - 12A^2 + 23A - 9I$

AKTU 2016, 2018, 2023

$$A^{-1} = \frac{1}{4} \begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & 1 \\ -1 & 1 & 3 \end{bmatrix}$$

$$5A - I = \begin{bmatrix} 9 & -5 & 5 \\ -5 & 9 & -5 \\ 5 & -5 & 9 \end{bmatrix}$$

Q.3 If $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ find A^{-1} and A^4 using Cayley-Hamilton theorem. Also show that for every integer

$$n \geq 3, A^n = A^{n-2} + A^2 - I$$

$$A^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ -1 & 1 & 0 \end{bmatrix}$$

$$A^4 = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}$$

Q.4 Using Cayley-Hamilton theorem, find the inverse of the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$ Also express the

polynomial $B = A^8 - 11A^7 - 4A^6 + A^5 + A^4 - 11A^3 - 3A^2 + 2A + I$ as quadratic polynomial in A and hence find B

$$A^{-1} = \begin{bmatrix} 1 & -3 & 2 \\ -3 & 3 & -1 \\ 2 & -1 & 0 \end{bmatrix}$$

AKTU 2015, 2019

$$B = A^2 + A + I = \begin{bmatrix} 16 & 27 & 34 \\ 27 & 50 & 61 \\ 34 & 61 & 77 \end{bmatrix}$$

Q.5 Express the polynomial $2A^5 - 3A^4 + A^2 - 4I$ as linear polynomial in A where $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$ AKTU 2017

$$138A - 403I$$

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Thank You